

Xyra

The Standalone Agentic IDE

Product, Architecture and Delivery Blueprint

Wiener Labs



This document specifies Xyra as an independent product built on permissively licensed foundations. It covers the product thesis, the licensing strategy that makes commercial distribution possible, the full system architecture, the context compression and acceleration protocols that differentiate the editor, the agent orchestration layer, a screen by screen and control by control interface specification, the user stories that justify each surface, the complete work breakdown with an issue register, and the generation prompts that turn this document into running code.

Confidential. Internal planning document.

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Chapter 1

Product Thesis

1.1 What Xyra is

Xyra is a desktop integrated development environment whose primary user is a developer working alongside autonomous agents. Every other editor treats the agent as a plugin bolted onto a text editing surface designed for a human typing character by character. Xyra inverts the relationship. The agent is a first class runtime with its own lifecycle, its own verification loop, its own memory, and its own presence in the interface. The text editor is one surface among several, and it is the surface a developer uses to inspect, correct and approve work that mostly originates elsewhere.

The product exists because three assumptions behind current agentic editors are wrong.

1. **Agents present unverified code.** A model writes a diff and the diff appears. Whether it compiles, whether the tests pass, whether it breaks a caller three modules away, all of that is discovered later by the human. Xyra refuses to display a change that has not been executed.
2. **Context is treated as a bucket.** Editors stuff files into a prompt until the window fills, then truncate. This wastes the majority of the budget on tokens that carry no decision weight. Xyra treats context as a compression problem with a measurable objective and spends its budget on the smallest representation that preserves the ability to act correctly.
3. **One session equals one task.** Work stops when the window fills, and the human restarts it. Xyra decomposes an objective into a dependency graph and runs each node in a fresh session, so total work is unbounded by any single context window.

1.2 The one sentence

Xyra is the IDE where an autonomous team of agents ships a project while you sleep, and every line they wrote was proven before you ever saw it.

1.3 Design principles

Proof before presentation.

No diff reaches a human surface until it has been executed against the project's own test or build command inside an isolated snapshot. This is enforced by the runtime, not by prompting.

Determinism where determinism is possible.

Import graphs, call graphs, symbol tables and dependency closures are computed, not inferred. Probabilistic retrieval is a fallback for questions that cannot be answered structurally, never the first resort.

Local by default.

Indexing, embedding, graph construction, compression, screenshot analysis, quality assurance runs and the entire mission state live on the developer's machine. Network calls are limited to the model provider the developer chose.

Every automated act is reversible and attributable.

One commit per unit of work, a decision journal that records why, and a time travel surface that can branch from any past decision.

The interface exposes the machine’s state, not a chat transcript.

What the agent is reading, what it decided, what it verified, what it is about to do, and what it cost. A transcript is the least informative possible view of an autonomous system.

Nothing blocks on a human unless a human must decide.

The supervisor asks nothing it can resolve. When a decision genuinely belongs to the developer, it arrives as a structured choice with costs attached, never as a terminal question.

1.4 Positioning

Capability	Mainstream agentic editors	Xyra
Verification of generated code	Human runs it after the fact	Runtime executes it in an isolated snapshot before display
Context assembly	Similarity search over chunks, truncation at the limit	Structural retrieval plus a compression pipeline with a token budget solver
Long horizon work	Bounded by one context window	Unbounded, ticket graph across fresh sessions
Visual correctness	Not modelled	Headless render, screenshot, vision verdict in the fix loop
Cross repository change	Manual, one repository at a time	Fleet impact analysis and sequential execution across repositories
Model relationship	Single vendor, metered	Multi vendor, adversarial review, flat rate capable
Failure handling	Session ends, human restarts	Retry, vendor swap, ticket split, quarantine, continue
State durability	Ephemeral chat	Durable mission state, decision journal, resumable after crash

Table 1.1: Positioning against the current generation of agentic editors.

1.5 Why a standalone product rather than a fork

The existing Xyra ships as a branded build of a GPL licensed editor. That path validated the product thesis quickly and it remains a legitimate distribution channel, but it imposes two permanent constraints. Source disclosure obligations follow every binary, which removes the option of a proprietary commercial edition. Upstream churn dictates the rebase schedule, which means engineering capacity is consumed by tracking another organisation’s decisions.

The standalone product removes both constraints by building on permissively licensed foundations only. Chapter 2 enumerates the verified license position of every dependency in the proposed stack. The strategic consequence is that the same codebase can be released as a free edition and licensed as a commercial edition without any restructuring.

1.6 Product surfaces

Xyra is one application with six primary surfaces. Each is specified control by control in Chapter 6.

Surface	Purpose
Workbench	Editing, navigation, diffing, terminal, source control. The classic IDE surface.
Mission Control	The autonomous supervisor: objective, ticket graph, progress, budget, controls.
Agent Theatre	Live orchestration of the agent team, lane by lane, with the artefacts each produced.
Context Inspector	What the agent is reading, weighted, with the ability to exclude and pin.
Verification Deck	Sandbox runs, visual checks, quality assurance sweeps, security and cost flags.
Atlas	Structural view of the codebase and the fleet: topology, impact, ownership, history.

Table 1.2: The six product surfaces.

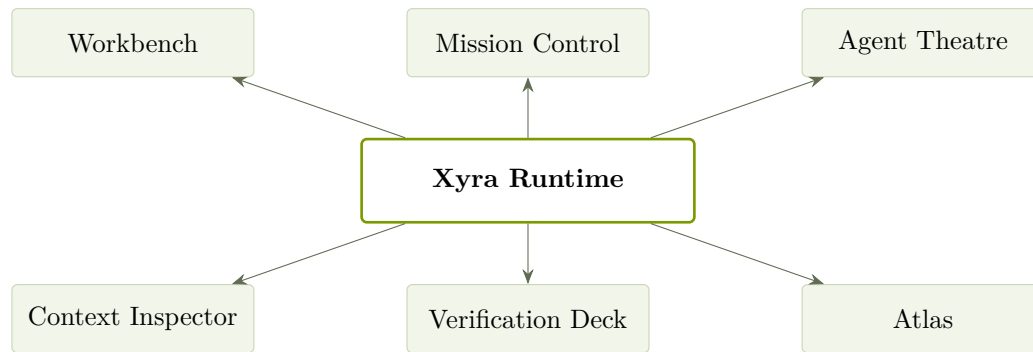


Figure 1.1: The runtime owns all state. Surfaces are projections of that state, never owners of it.

1.7 Non goals

Stating what Xyra does not attempt is as load bearing as stating what it does.

- Xyra is not a general purpose text editor competing on startup time for editing a single configuration file. It is a project scale environment.
- Xyra does not host models. It orchestrates model providers the developer already pays for, including local ones.
- Xyra does not attempt to replace the shell or the browser. It drives them and reads their output.
- Xyra does not ship an interactive debugger in the first product generation. Runtime evidence reaches the agent through test output, traces and logs rather than through a breakpoint session, which is why no debug adapter appears in the issue register and no breakpoint control appears in the interface specification.
- Xyra does not implement its own version control. It drives git.
- Xyra does not implement real time collaborative editing. Multiple developers work through version control, and the collaboration surface in the title bar is reserved for a later generation.
- Xyra does not target mobile or web delivery in the first product generation.

The commercial model, pricing and edition split are deliberately outside the scope of this document. Chapter 2 establishes the licensing position that keeps every commercial option open, which is the only part of that decision that constrains engineering.

Chapter 2

Open Source Foundation and Licensing

2.1 The commercial constraint

Every dependency in Xyra must satisfy one rule: it must permit distribution of a binary whose corresponding source is not disclosed. In practice this means MIT, Apache 2.0, BSD, ISC, Zlib, Unlicense or CC0. It excludes GPL and LGPL in linked form, and it excludes MPL 2.0 as well, for the reason given below.

This rule is what separates the standalone product from the current fork. It is not a philosophical preference, it is the precondition for a commercial edition.

The allow list is exactly MIT, Apache-2.0, BSD-2-Clause, BSD-3-Clause, ISC, Zlib, Unlicense, CC0-1.0 and Unicode-3.0. The last entry is not a concession: the Unicode license carries no copyleft, and the internationalisation family that arrives through url and idna is unavoidable in any project that parses a link or normalises text. MPL-2.0 is **not** on it. Its copyleft is file scoped rather than project scoped, which makes it survivable in principle, but a per file disclosure obligation inside a commercial binary is a compliance burden with no compensating benefit when a permissive alternative exists. A dependency outside the list enters only through an explicit entry in the exception list inside the license check script, which puts the decision in front of a reviewer rather than inside a transitive dependency nobody read.

2.2 Verified license position

The table below was compiled from the crates.io registry metadata and from the license declarations inside each project, rather than from documentation or memory. Every entry was checked while writing this document.

Component	License	Role in Xyra
<i>Interface and rendering</i>		
gpui	Apache-2.0	GPU accelerated UI framework, element tree, layout, input
wgpu	MIT OR Apache-2.0	Graphics abstraction, fallback rendering path
vello	Apache-2.0 OR MIT	Vector rendering, alternative compositor path
cosmic-text	MIT OR Apache-2.0	Text shaping and layout
swash	Apache-2.0 OR MIT	Font parsing, glyph rasterisation
<i>Editor core</i>		
ropey	MIT OR Apache-2.0	Rope data structure for the text buffer
tree-sitter	MIT	Incremental parsing, concrete syntax trees

Component	License	Role in Xyra
tree-sitter-highlight	MIT	Syntax highlighting over the parse tree
similar	Apache-2.0	Text diffing for review surfaces
imara-diff	Apache-2.0	High performance line diffing for large files
<i>Language intelligence</i>		
lsp-types	MIT	Language server protocol type definitions
tower-lsp or hand rolled client	MIT	Language server transport and lifecycle
<i>Project and system</i>		
gix (gitoxide)	MIT OR Apache-2.0	Pure Rust git implementation
ignore	Unlicense OR MIT	Gitignore aware directory traversal
globset	Unlicense OR MIT	Glob matching for path rules
notify	CC0-1.0	Filesystem watching
alacritty_terminal	Apache-2.0	Terminal emulation backend
tokio	MIT	Asynchronous runtime for the service layer
dashmap	MIT	Concurrent maps for shared indices
<i>Retrieval and storage</i>		
tantivy	MIT	Full text and structural search index
usearch	Apache-2.0	Approximate nearest neighbour vector index
fastembed	Apache-2.0	Local embedding generation
candle-core	MIT OR Apache-2.0	Local model execution for embeddings and rerankers
tiktoken-rs	MIT	Token counting for budget accounting
redb	MIT OR Apache-2.0	Embedded key value store for the durable state
rusqlite	MIT	Relational store for the graph and journal
zstd	MIT	Compression of context artefacts and caches
<i>Agent integration</i>		
agent-client-protocol	Apache-2.0	Protocol for external agent processes
rmcp	Apache-2.0	Model context protocol client and server
wasmtime	Apache-2.0 WITH LLVM-exception	Sandboxed extension execution
<i>Interface support</i>		
fuzzy-matcher	MIT	Fuzzy matching for pickers and the palette

Component	License	Role in Xyra
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Table 2.1: Verified license position of the proposed dependency set.

2.3 What this table means

Two conclusions follow directly.

First, **a complete IDE can be assembled without touching a single copyleft dependency**. The UI framework, the rope, the parser, the terminal, the git implementation, the language server client, the vector index and the extension sandbox are all permissive. There is no component in the critical path that forces source disclosure.

Second, **the highest leverage reuse is gpui**. It is Apache 2.0, it is production proven in a shipping editor, and it solves the single hardest problem in this project, which is a GPU accelerated UI framework with correct text rendering and input handling on three platforms. Writing that from scratch would consume the majority of the engineering budget and produce a worse result. Using it is not a compromise on independence, because Apache 2.0 imposes no disclosure obligation and no upstream coupling beyond the version we pin.

2.4 Dependency strategy

Pin the UI framework to its published release.

gpui is taken from the registry at an exact published version under Apache 2.0, not from the upstream editor's source tree. This distinction is load bearing and was discovered while implementing it: in that tree the path `gpui` to `sum_tree` to `ztracing` reaches GPL-3.0-or-later code, which would put a disclosure obligation on this binary and defeat the reason the standalone product exists. The published crate does not carry that edge. Upstream versions are adopted deliberately, never automatically, which removes the rebase treadmill that characterises the fork.

Judge a dependency by whether it ships.

A build dependency executes during compilation and never links into the artefact, so its obligations do not follow the binary. The gate records such cases as named exceptions carrying their reason rather than widening the allow list for everything.

Wrap every third party boundary.

Editor code never calls gpui directly. It calls `xyra-ui`, a thin internal crate that re-exports the element model we actually use. The same applies to the parser, the git layer and the terminal. The cost is one indirection and the benefit is that any dependency can be replaced without touching product code.

Prefer boring dependencies.

Fewer, larger, well maintained crates over many small ones. Every dependency is a license question, a supply chain question and a maintenance question.

No network dependency at build time.

The build must succeed from a vendored source tree with no registry access, which is a hard requirement for reproducibility and for enterprise customers with air gapped environments.

2.5 License hygiene in the repository

- A machine readable manifest at `licenses/manifest.toml` lists every dependency, its license, its source and the reason it is present.
- A continuous integration job fails the build when a dependency appears whose license is not on the allow list, using cargo-deny or an equivalent check.
- Vendored third party source lives under `third_party/` with its original license file intact and unmodified.
- The notice file shipped with the product aggregates every attribution required by the Apache 2.0 and MIT terms.

2.6 Reference implementations worth studying

These projects are not dependencies. They are prior art whose architecture repays study before writing the equivalent subsystem. All are permissively licensed, so reading them carries no contamination risk, and porting an approach from them is legally clean when the license is respected and attribution given.

Project	License	What to learn from it
Helix	MPL-2.0	Multi cursor and selection model, tree-sitter integration depth
Lapce	Apache-2.0	Rope plus WASM plugin boundary, multi window process model
Zed gpui	Apache-2.0	Element tree, layout engine, GPU text pipeline
Alacritty	Apache-2.0	Terminal grid, escape sequence handling, performance discipline
Tantivy	MIT	Index segment design, query planning
Wasmtime	Apache-2.0	Capability based sandboxing for untrusted extensions
Ripgrep and ignore	MIT	Directory traversal that respects ignore semantics at speed

Table 2.2: Permissively licensed prior art.

Chapter 3

System Architecture

3.1 Layering

Xyra is organised as five layers. Each layer may call downward and emit events upward. No layer may call upward directly. This rule is what keeps the surfaces replaceable and the core testable without a display.

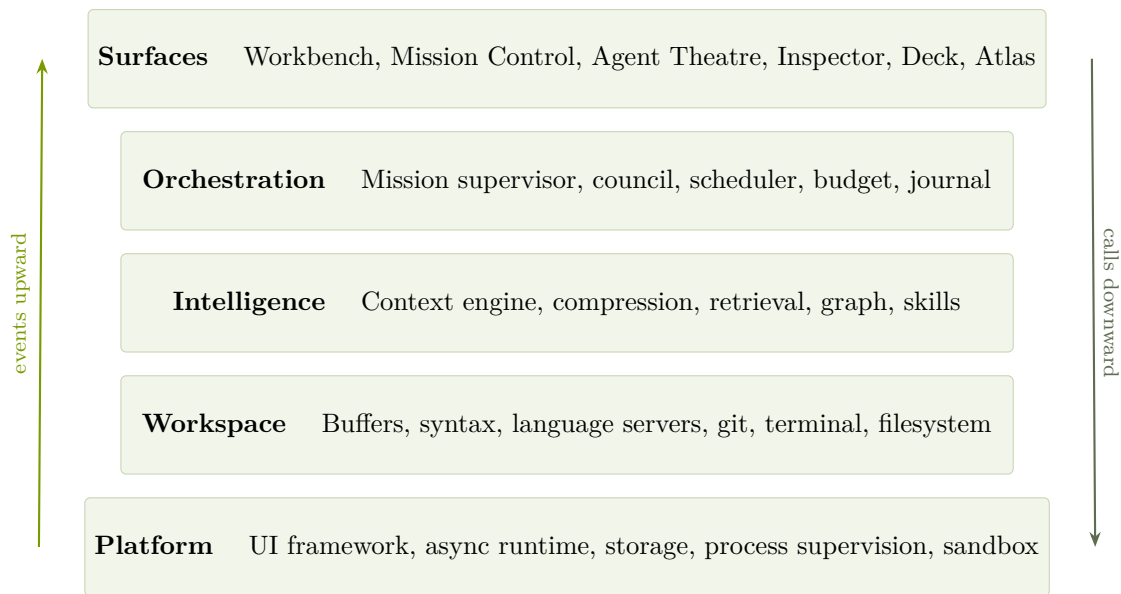


Figure 3.1: Five layer architecture with a strict dependency direction.

3.2 Process model

Xyra runs as a small constellation of processes rather than one monolith. The reason is containment: an agent process that hangs, a language server that leaks, or an extension that panics must never take down the editor, and must never be able to block the interface thread.

xyra

The interface process. Owns the window, the element tree, input handling and rendering. Holds no authoritative state; it renders a projection pushed from the core and sends intents back.

xyrad

The core service. Owns buffers, the project model, the context engine, the mission supervisor, the journal and all persistence. Survives interface restarts, which means a crashed window never loses a running mission.

Language servers

One child per language per workspace, supervised with restart backoff and a health probe.

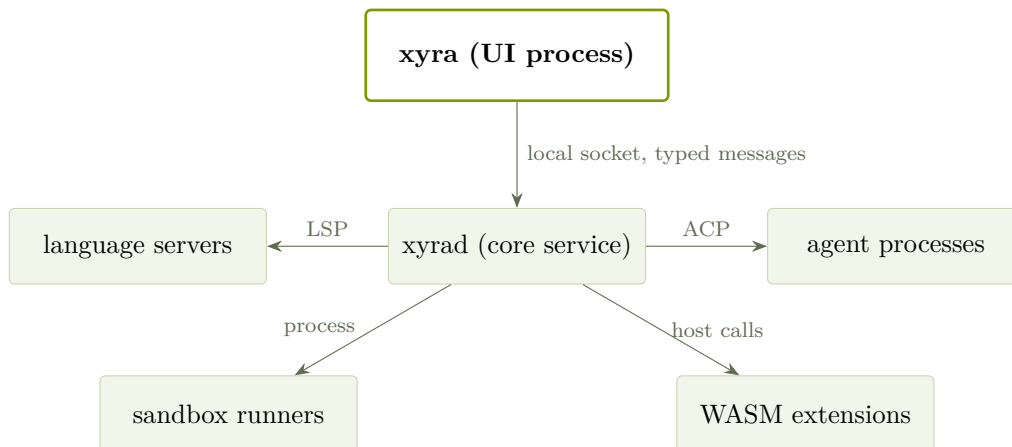


Figure 3.2: Process topology. The interface never speaks to an agent or a language server directly.

Agent processes

External agent binaries speaking the agent client protocol over stdio. Each runs in its own working directory with an environment the core constructs.

Sandbox runners

Ephemeral processes executing tests inside an isolated worktree. Killed on budget expiry without affecting anything else.

WASM extensions

Untrusted third party code with an explicit capability list, no ambient filesystem access and no network unless granted.

3.3 Crate topology

Crate	Kind	Responsibility
xyra-ui	library	Element model wrapper over the vendored UI framework, theme tokens, primitives
xyra-text	library	Rope buffer, transactions, undo history, multi cursor selection algebra
xyra-syntax	library	Parser host, incremental reparse, highlight queries, structural selection
xyra-lang	library	Language server client, capability negotiation, diagnostics, completion
xyra-project	library	Worktree model, ignore semantics, file watching, path resolution
xyra-vcs	library	Git operations, status, diff, blame, staging, commit, branch
xyra-term	library	Terminal grid, PTY supervision, escape parsing, search
xyra-context	library	Symbol and import graph, retrieval, compression pipeline, budget solver
xyra-memory	library	Durable stores, caches, journal, mission state, migrations
xyra-agent	library	Agent process supervision, protocol clients, tool exposure, transcript capture
xyra-mission	library	Ticket graph, supervisor loop, failure policy, budgets

Crate	Kind	Responsibility
xyra-verify	library	Sandbox execution, visual verification, quality assurance sweeps, flags
xyra-fleet	library	Multi repository registry, cross repository search and impact
xyra-skills	library	Skill registry, resolution, injection, precedence
xyra-ext	library	Extension host, capability enforcement, manifest validation
xyra-proto	library	Message types shared between the interface and the core
xyrad	binary	Core service binary, wiring, supervision tree
xyra	binary	Interface binary, surfaces, keymap, command registry

Table 3.1: Crate responsibilities. One crate, one reason to change.

3.4 State ownership and the projection model

The interface holds no authoritative state. Every surface receives a projection: a serialisable snapshot of exactly what it needs to render, versioned so that stale frames are discarded. User actions travel back as intents, never as mutations.

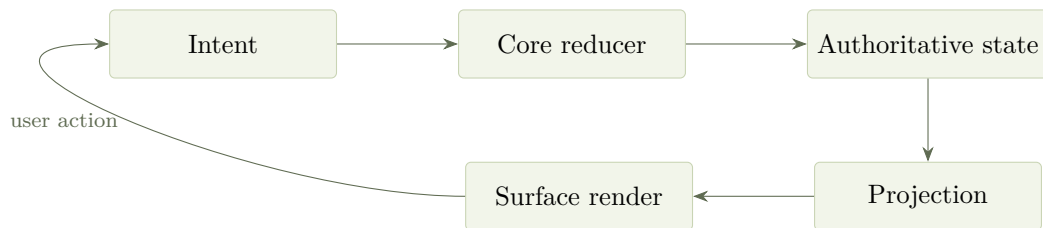


Figure 3.3: Unidirectional flow. The loop is closed only by user action.

This is what makes the product testable. Every surface can be exercised by feeding projections and asserting on the intents produced, with no window open and no GPU present.

3.5 Concurrency model

- The interface thread never blocks. Any operation with unbounded duration is a task on the core's runtime, and the interface shows an intermediate state.
- Buffer edits are applied on a single owning task per buffer. Concurrent edits arrive as transactions and are ordered, never merged optimistically.
- Indexing, embedding and graph construction run on a bounded worker pool sized to physical cores minus two, so the machine stays usable.
- Every long running task carries a cancellation token and a budget. A task without a budget is a defect.
- Agent sessions are supervised with a timeout, an inactivity watchdog and a hard kill path, because the dominant real world failure is a subprocess that never exits.

Store	Technology	Contents
Project index	SQLite	Symbols, import edges, file hashes, chunk metadata
Vector index	Embedded ANN file	Chunk embeddings, quantised, memory mapped
Mission state	JSON with atomic replace	Ticket graph, attempts, budgets, commits
Journal	Append only JSON lines	Decisions, verdicts, verification results
Caches	Content addressed files	Compression artefacts, screenshots, transcripts
Settings	TOML	User and workspace configuration, layered

Table 3.2: Storage responsibilities. Nothing important lives only in memory.

3.6 Persistence

Every store is per workspace under `.xyra/` except global caches, which live under the platform cache directory. Deleting `.xyra/` must degrade the product to a cold start and never corrupt the repository.

3.7 Failure containment

Failure	Containment
Language server crash	Supervised restart with exponential backoff, diagnostics preserved, editing unaffected
Agent process hang	Inactivity watchdog kills the session, ticket is marked failed, supervisor continues
Extension panic	WASM trap is caught, extension is disabled for the session, a flag is raised
Core service crash	Interface shows a reconnect state, mission resumes from durable state on restart
Interface crash	Core keeps working, mission continues, window reopens onto live state
Disk full	Writes fail closed, mission halts with a stated reason, nothing is half written
Corrupt index	Detected by checksum, index rebuilt in background, structural queries degrade to search

Table 3.3: Every failure has a named containment strategy.

Chapter 4

The Context Engine

4.1 The objective function

Every agentic editor faces the same problem and almost none of them state it as an optimisation. Given a token budget B , choose a set of context items S that maximises the probability the agent produces a correct action.

$$\max_S P(\text{correct action} \mid S) \quad \text{subject to} \quad \sum_{i \in S} \text{tokens}(i) \leq B$$

The naive implementation ranks chunks by embedding similarity and fills the budget until it overflows. This is wrong in three distinct ways. It ignores that some items are load bearing regardless of similarity, such as the exact signature of the function being modified. It ignores that many high similarity items are redundant with each other. And it pays full token price for information that could be represented at a fraction of the cost without losing decision value.

Xyra treats this as a compression problem with a value model. The engine is built from four mechanisms: structural representation, tiered escalation, delta encoding, and a budget solver. Around them sit accelerators that reduce latency and cost without changing the selected set.

4.2 Structural Context Protocol

The Structural Context Protocol, abbreviated SCP, is the canonical representation of a source file inside Xyra. It is not the file text. It is a projection of the file's parse tree into a form that preserves everything an agent needs to reason about the file's interface and dependencies, while discarding the bodies it does not need.

An SCP record for a file contains the module path and language, the ordered list of exported and internal symbols with their full signatures, type declarations in full, documentation comments attached to symbols, the import list resolved to concrete paths, the outbound call edges by symbol, and a stable content hash per symbol body so a body can be requested later by handle.

Representation	Tokens	Ratio	Preserves
Raw file text	3200	1.00	Everything
SCP level 2	980	0.31	Signatures, types, docs, imports, call edges, body handles
SCP level 1	420	0.13	Signatures, types, imports
SCP level 0	130	0.04	Module path, exported symbol names, import list

Table 4.1: Indicative compression profile for a typical eight hundred line source file. Ratios are representative of the design target, and the harness in Chapter 9 measures the real distribution per language.

The critical property is that SCP is **lossless with respect to the interface**. An agent reading SCP level 2 knows every symbol it can call, with exact parameter types and return types. It does not know how those symbols are implemented, and in the majority of tasks it does not need to. When it does, it asks.

4.3 Tiered escalation

Context is assembled as a ladder, and the agent climbs it only where it must. This inverts the usual flow, in which the editor guesses what the agent needs before the agent has reasoned about the task.

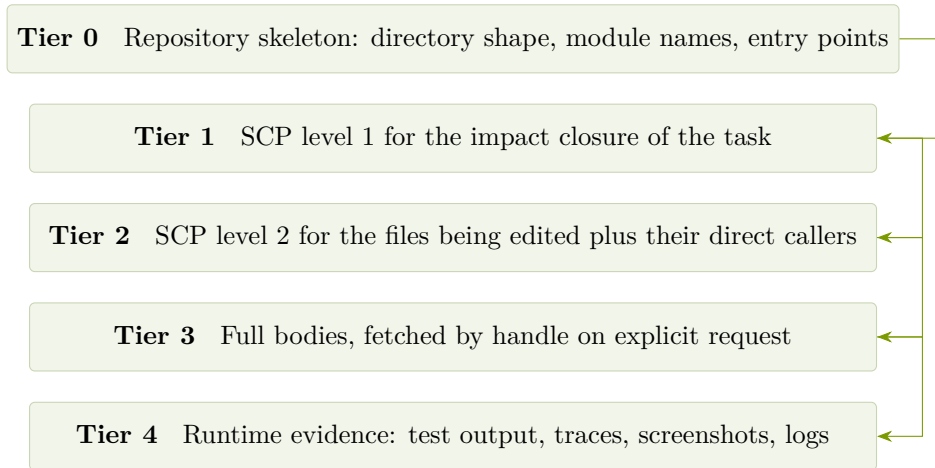


Figure 4.1: The context ladder. Escalation is driven by the agent’s own tool calls, not by the editor’s guess.

Tier 0 and Tier 1 are pushed into the session automatically because they are cheap and always relevant. Tier 2 is computed from the deterministic impact closure described in Section 4.4. Tier 3 and Tier 4 are pulled by the agent through tools, which means the cost is paid only when the information is actually wanted.

4.4 Structural retrieval

Before any embedding is consulted, Xyra answers the question structurally.

Import graph.

Parsed from source, resolved to concrete files, stored as edges. Gives the exact set of files that would break if a module changes, by transitive distance.

Symbol table.

Every definition with its kind, location and visibility, so a symbol name resolves to a definition without similarity search.

Call edges.

Outbound calls per symbol, which produces a call closure for a change to a single function rather than a whole file.

Test mapping.

Which test files exercise which source files, derived from imports and from execution traces when available. This makes the verification loop targeted rather than running the whole suite each time.

Blame and history.

Which commits last touched the region, which gives an agent the reason a line exists before it deletes it.

Embedding search is consulted for a narrow class of questions: locating behaviour described in prose when the developer does not know the symbol name. It is a fallback, and the interface labels its results as such so the developer can see when the machine is guessing.

4.5 Semantic delta encoding

Within a mission, consecutive sessions overlap heavily. Re-sending the same skeleton on every ticket wastes a large fraction of the budget. Xyra addresses this with a delta protocol.

The core maintains, per mission, a content addressed dictionary of context items already transmitted. When assembling the next session, items already in the dictionary are referenced by handle rather than repeated, and the session preamble carries only the delta: what changed since the last transmission, plus the handles for everything unchanged.

Element	Encoding	Cost
Unchanged file, previously sent	Handle reference with symbol list	Tens of tokens
Changed file	Delta against the previous SCP record	Proportional to the change
New file in scope	Full SCP record at the required tier	Full record cost
Removed from scope	Explicit eviction notice	Negligible

Table 4.2: Delta encoding between sessions in a mission.

Because each ticket runs in a fresh session, the dictionary is not a context window trick. It is a compact resumption record that lets a new session reach the same understanding as its predecessor for a fraction of the tokens.

4.6 The budget solver

Given candidate items with an estimated value and a measured token cost, selection is a knapsack problem with dependencies and redundancy penalties.

- **Value** combines structural proximity to the task target, recency of change, authorship of the failing test, and explicit developer pinning. Similarity contributes, but it never dominates a structural signal.
- **Cost** is measured with a real tokeniser for the target model, not estimated by character count. Estimating by characters is wrong by up to forty percent on code.
- **Redundancy** is penalised. Two chunks with near identical content contribute once.
- **Dependencies** are respected. Including a symbol implies including the types in its signature.
- **Floors** are enforced. The file being edited is always at Tier 2 regardless of the solver’s opinion, because omitting it produces confident nonsense.

The solver runs greedily by value density with a repair pass, which is fast enough to run on every assembly and close enough to optimal that the difference does not matter in practice.

4.7 Accelerators

Compression reduces cost. Accelerators reduce latency and repeated work. They do not change which items are selected, which keeps them safe to enable by default.

Accelerator	Mechanism
Stable prefix ordering	Context is emitted in a canonical order with the invariant material first, so provider side prefix caches hit across sessions instead of missing on reordering.
Speculative prefetch	While the agent is reasoning, the graph predicts the next files it will request and warms their SCP records and bodies.
Incremental embedding	Content hashed per chunk, so only changed chunks are re-embedded. A file rename costs nothing.
Quantised vector index	Embeddings stored quantised and memory mapped, which keeps a large repository index resident without dominating memory.
Local reranking	A small cross encoder run locally reorders the candidate set before the budget solver, which improves selection quality without a network call.
Parallel lens execution	Independent review lenses run concurrently rather than in sequence, so an adversarial review costs one lens of latency rather than three.
Warm sandbox worktrees	A pool of prepared worktrees with dependencies already installed, so verification does not pay the setup cost on every ticket.
Test selection	The test mapping restricts verification to the tests that can be affected, with a full suite run at ticket boundaries and before completion.
Streaming to disk	Agent output is persisted as it arrives, so a crashed session is resumable and a long transcript never lives only in memory.
Compression of artefacts	Transcripts, screenshots and traces are stored compressed and content addressed, deduplicated across missions.

Table 4.3: Latency and cost accelerators.

4.8 Context expansion integrations

Structural knowledge of the repository is necessary but not sufficient. A large class of agent errors comes from information that simply is not in the code. These integrations bring that information into the ladder as first class context, each behind an explicit capability so a developer can see and control what leaves the machine.

Integration	Brings in	Answers the question
Language server	Hover types, call hierarchy, references, definitions	What is the real type here, who calls this
Version history	Blame, commit messages, pull request bodies	Why does this line exist
Issue tracker	Linked issues, acceptance criteria	What was actually asked for
Documentation fetch	Current library documentation by version	What is the correct API today, not at training time
Runtime traces	Execution paths, values at failure	What actually happened, not what should have
Test results	Failure output, flake history	Which failures are real

Integration	Brings in	Answers the question
Design assets	Exported mockups, design tokens	What is this supposed to look like
Schema and migrations	Database schema, pending migrations	What shape is the data
Observability	Error rates, slow queries, recent incidents	What is breaking in production
Dependency registry	Package sizes, versions, advisories	What does adding this cost
Fleet siblings	Contracts consumed by other repositories	Who else breaks if this changes
Terminal history	Recent commands and their output	What has the developer already tried

Table 4.4: Context expansion integrations and the question each one closes.

4.9 The assembly pipeline

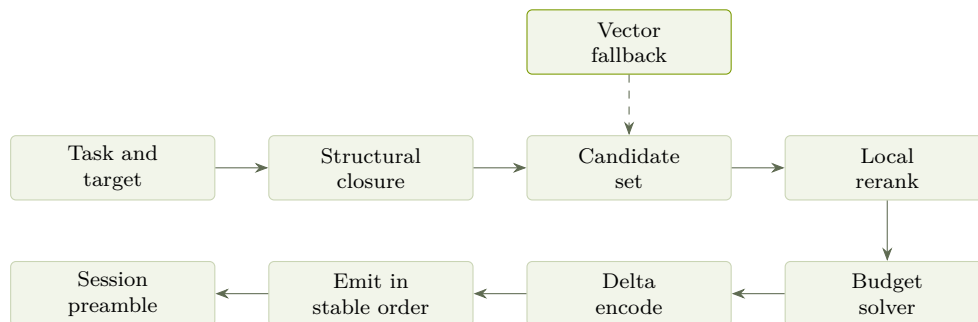


Figure 4.2: Context assembly. Similarity search enters only as a labelled fallback.

4.10 Measurement

None of this is worth shipping unless it is measured. The context engine carries a permanent evaluation harness with four metrics, reported per mission and visible in the Context Inspector.

- **Budget utilisation.** Tokens spent versus tokens available, and the fraction spent on each tier.
- **Compression ratio.** Tokens emitted versus tokens the naive representation would have required for the same item set.
- **Escalation rate.** How often the agent had to climb to Tier 3, which is the signal that the lower tiers are under specified.
- **First attempt success.** The fraction of tickets whose first session passes verification, which is the only end to end measure that matters.

Chapter 5

The Agent Layer

5.1 Roles rather than one assistant

A single general assistant is the wrong abstraction for autonomous work, because the prompting, the tool set and the failure handling that make an implementer effective make a reviewer credulous. Xyra runs a team of specialised roles with distinct mandates and distinct tool grants.

Role	Mandate	Write access
Planner	Decompose an objective into a dependency ordered ticket graph	None
Architect	Produce and defend a design, challenge another design	None
Implementer	Complete exactly one ticket, leave tests passing	Worktree
Reviewer	Cross examine a diff through one lens, return a structured verdict	None
Verifier	Execute tests, render pages, run quality sweeps, report facts	Sandbox only
Repairer	Fix a specific named failure inside a snapshot	Snapshot
Replanner	Split a repeatedly failing ticket into smaller tickets	None
Scribe	Summarise, journal, produce release notes and reports	Documents

Table 5.1: Agent roles. Write access is a property of the role, not of the model.

Visual verification, quality sweeps and cross repository analysis are capabilities of the Verifier role rather than separate roles, which keeps the role set closed at eight. The Agent Theatre in Chapter 6 renders exactly one lane per role in this table.

The vendor behind each role is configurable, and the default deliberately places the reviewer on a different vendor from the implementer. A model reviewing its own output shares its own blind spots.

5.2 The mission supervisor

The supervisor is the component that makes unattended work possible. It owns a durable ticket graph and drives it to completion without asking the developer anything it can resolve itself.

5.2.1 Why fresh sessions

The central design decision is that each ticket runs in a session that has never seen any other ticket. This is counterintuitive until the arithmetic is done. A session that accumulates context degrades in two ways: it fills the window, and it accumulates the residue of earlier reasoning, including earlier mistakes. A fresh session with a precise brief and a compressed structural context reaches higher first attempt success than a long running session at ticket fifty, and the total work a mission can perform stops being bounded by any window at all.

5.2.2 Durability

Mission state is written atomically after every state transition. The consequences are concrete: closing the laptop pauses a mission, a crash loses at most the ticket in flight, and resuming is a single command that reads the graph and continues. The state file is the mission; the process is disposable.

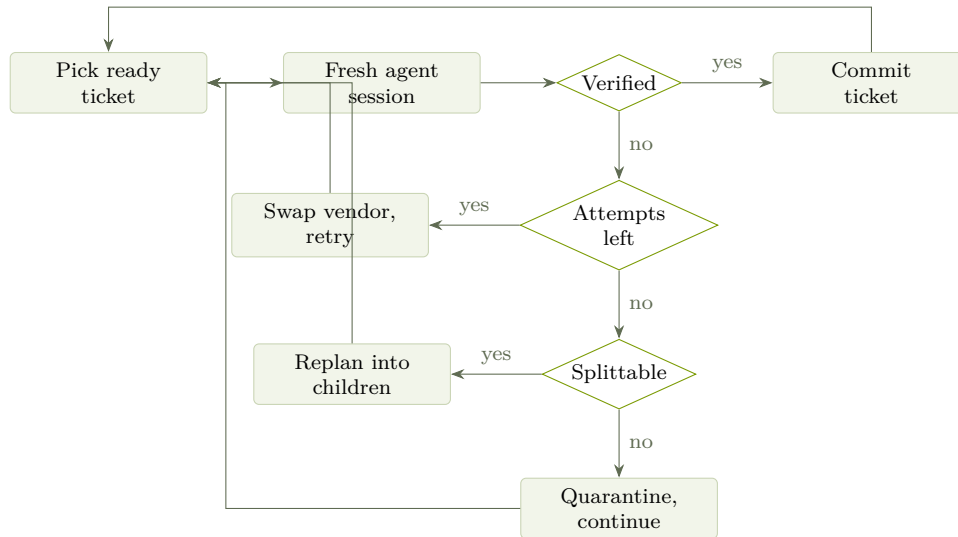


Figure 5.1: Supervisor loop. Failure is a policy with four escalations, never a stop.

5.2.3 Budgets and brakes

Autonomy without limits is a liability. Every mission carries hard caps on total sessions, wall clock duration, attempts per ticket and consecutive failures, plus a stop flag the developer can set at any moment which halts the loop after the ticket in flight completes. Exceeding any cap halts the mission with a stated reason rather than continuing on hope.

5.3 Verification

5.3.1 Isolated execution

Verification never runs in the developer’s working tree. The supervisor snapshots the uncommitted change set into a throwaway worktree and executes the project’s own test or build command there. The developer’s tree is untouched, which means verification can run while the developer keeps working, and a failed attempt cannot leave debris behind.

5.3.2 Visual verification

For any change with a reachable page, a verifier renders it headless, captures a screenshot and asks a vision capable model whether the result matches the stated intent. The verdict is structured, with issues ranked by severity, which is what makes it usable inside an automatic fix loop rather than as decoration. A design export can be supplied, in which case the comparison is against the intended design rather than against a description.

5.3.3 Adversarial review

The council runs a diff past a rival vendor through independent lenses, by default correctness, security and house conventions, executed in parallel. Each lens returns findings with severity, and a policy decides whether the aggregate blocks. Policy is per repository and per path, so code that touches money or credentials can demand more lenses and block on lower severities than code that touches a marketing page.

5.3.4 Quality assurance sweeps

A verifier drives the running application as a hostile user: crawling same origin pages, filling every input with type aware adversarial values, clicking aggressively, and collecting console errors, uncaught exceptions, server errors, failed requests, accessibility problems and load timings. Findings are deduplicated and ranked, then fed back as tickets rather than delivered as a raw log.

5.4 Skills

Skills are the mechanism by which a team encodes its own standards so that every agent session inherits them without being told.

Skill	Encodes
Autonomy	Prove before presenting: verify, look at the result, analyse impact before touching shared contracts
Terminal discipline	Never run an interactive command or a process that does not exit, per tool flags tabulated
Conventions	House rules for code, commits and interface copy
Review	The checklist a reviewer applies, including domain specific traps
Domain safety	Rules for money arithmetic, credentials, and any high consequence domain
Shipping	The gate before a change leaves the machine
Prompt discipline	Turning a vague instruction into a scoped one before acting

Table 5.2: The default skill set. Teams add their own.

Skills resolve in a precedence chain: workspace, then user, then bundled defaults. A workspace skill overrides a user skill of the same name, which lets a team enforce a standard on a repository without touching anyone’s personal configuration. The Context Inspector shows exactly which skills were injected into any given session, because an invisible instruction that changes behaviour is a debugging nightmare.

5.5 Fleet operations

A contract rarely lives in one repository. Fleet raises every structural capability from the repository to the group.

- Repositories are registered from the developer’s version control provider, not from a hand written file, and missing ones are cloned on registration.
- Cross repository search resolves a symbol or endpoint name across every member.
- Cross repository impact separates the repository that defines a contract from those that consume it, and orders the resulting work accordingly.
- A fleet refactor executes the same intent through every affected repository in dependency order, verifying each one independently and preparing linked changes.

5.6 Tool surface exposed to agents

Agents act through a bounded tool set. Every tool is auditable, every tool is capability gated, and no tool grants ambient authority.

Tool	Contract
read_file	Returns a file at a requested SCP tier or in full, by handle
search_code	Structural search first, similarity fallback clearly labelled
impact	Deterministic closure for a symbol or file, by transitive distance
graph	Neighbourhood of a file: imports, importers, defined symbols
edit	Applies a transaction to a buffer, always reversible
run_tests	Executes the project's test command inside a snapshot
verify_patch	Full verification of the current change set, returns a verdict
render_check	Renders a page and returns a structured visual verdict
compare_design	Renders a page and compares it against a design export
qa_sweep	Drives the running application adversarially, returns ranked defects
fleet_search	Cross repository search
fleet_impact	Cross repository impact analysis
journal	Records a decision with its rationale against the current commit
ask_developer	Escalates a genuine decision as a structured choice with costs

Table 5.3: Agent tool surface. Anything not on this list is not reachable.

5.7 Escalation to the developer

The supervisor escalates only when a decision genuinely belongs to the human: an architectural fork with real trade offs, an ambiguity in the objective that changes the outcome, or an action with consequences outside the repository. Escalation is never a terminal question. It is a structured choice: each option carries its measured cost, its dependency weight and a sketch of the resulting structure, and the mission continues on unblocked tickets while the choice is pending.

Chapter 6

Interface Specification

This chapter specifies every surface, every region and every control. Each control carries a stable identifier used by the implementation, the automated interface tests and the accessibility layer. A control that is not in this chapter does not ship.

6.1 Design system

6.1.1 Tokens

Token	Value	Applied to
space.1 to space.8	2, 4, 6, 8, 12, 16, 24, 32	All padding and gaps, no arbitrary values
radius.sm / md / lg	4, 6, 10	Controls, cards, panels
control.height.sm	28	Compact toolbar controls
control.height.md	34	Default buttons, inputs
control.height.lg	42	Primary actions, empty state actions
text.xs to text.xl	11, 12, 13, 15, 19	Labels, body, headings
icon.sm / md / lg	14, 16, 20	Inline, control, feature
elevation.0 to 3	Flat, subtle, raised, floating	Panels, popovers, dialogs, drag

Table 6.1: Design tokens. Sizes are in logical pixels at a scale factor of one.

6.1.2 Rules that are not negotiable

- Every control carries a visible text label unless it is in a dense toolbar, and even then it carries an accessible name and a tooltip. Icon only interfaces are a discoverability failure.
- Minimum interactive target is thirty four logical pixels in the primary flow.
- Sentence case everywhere. No capitalised words in interface copy.
- Every destructive action is either reversible or confirmed, never both silent and permanent.
- Every long running action shows determinate progress when a total is known, and an elapsed counter when it is not.
- Colour never carries meaning alone. Severity is colour plus a glyph plus a word.

6.2 Window anatomy

6.3 Title bar

Control	Label	Behaviour
ctl.title.project	Project name	Opens the recent projects menu
ctl.title.branch	Branch name with dirty dot	Opens branch switcher with search

Control	Label	Behaviour
ctl.title.mission	Mission pill, for example 7/19	Present only during a mission. Click focuses Mission Control. Shows a determinate ring
ctl.title.model	Active model and role routing	Opens the routing popover: implementer, reviewer, planner, embedding
ctl.title.share	Share	Opens the collaboration menu when enabled
ctl.title.zoom	Window controls	Platform native

Table 6.2: Title bar controls.

6.4 Activity bar

Vertical, icon plus short label, ordered and reorderable. Each entry toggles a dock panel and shows a badge when it has unseen state. Five entries open the primary surfaces named in Chapter 1; the remainder open supporting panels that live inside the Workbench surface rather than constituting surfaces of their own.

Control	Label	Badge	Opens
ctl.activity.explorer	Files	Modified count	Explorer
ctl.activity.search	Search	Match count	Search
ctl.activity.vcs	Git	Changed count	Source control
ctl.activity.mission	Mission	Progress ring	Mission Control
ctl.activity.agents	Agents	Active lanes	Agent Theatre
ctl.activity.context	Context	Budget percent	Context Inspector
ctl.activity.verify	Verify	Failing count	Verification Deck
ctl.activity.atlas	Atlas	None	Atlas
ctl.activity.ext	Extensions	Update count	Extensions

Table 6.3: Activity bar entries.

6.5 Workbench: the editor surface

6.5.1 Tab strip

Control	Behaviour
ctl.tab.item	File name, dirty dot, diagnostic underline, close on middle click
ctl.tab.agentmark	A small marker on tabs whose content was written by an agent and not yet reviewed by a human
ctl.tab.pin	Pin, keeps the tab through bulk close
ctl.tab.split	Split right, split down
ctl.tab.overflow	Overflow list with search when tabs exceed the width
ctl.tab.newfile	New file in this group

Table 6.4: Tab strip controls.

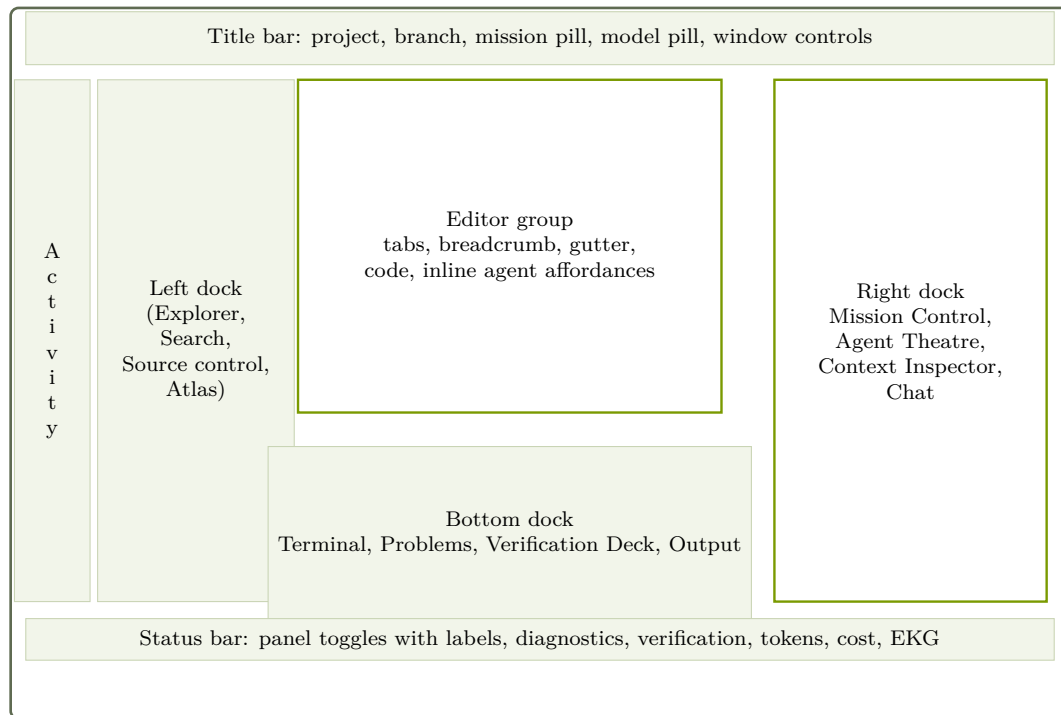


Figure 6.1: Window regions. The right dock is where autonomous work is observed and steered.

6.5.2 Gutter and inline affordances

Control	Behaviour
ctl.gutter.line	Line numbers, relative mode optional
ctl.gutter.fold	Fold and unfold by syntax node
ctl.gutter.vcs	Added, modified and removed markers, click opens the inline diff
ctl.gutter.blame	Inline blame on hover, showing the commit and its message
ctl.gutter.agent	Marker on ranges an agent changed in the current mission, with the ticket identifier
ctl.inline.ask	Inline agent prompt at the cursor, opened with a shortcut
ctl.inline.explain	Explain this symbol, uses structural context rather than a raw dump
ctl.inline.impact	Show impact for the symbol under the cursor, opens the closure in Atlas
ctl.inline.verify	Verify this file now, runs mapped tests only
ctl.inline.accept	Accept the agent range, clears the marker
ctl.inline.revert	Revert the agent range to the pre mission state

Table 6.5: Gutter and inline controls.

6.5.3 Breadcrumb and minimap

- `ctl.breadcrumb.segment` navigates by structural path, module then type then function, each segment opening a filtered symbol list.
- `ctl.minimap.toggle` switches the minimap between off, code and heat, where heat colours the map by how much agent attention each region received in the current mission.

6.6 Mission Control

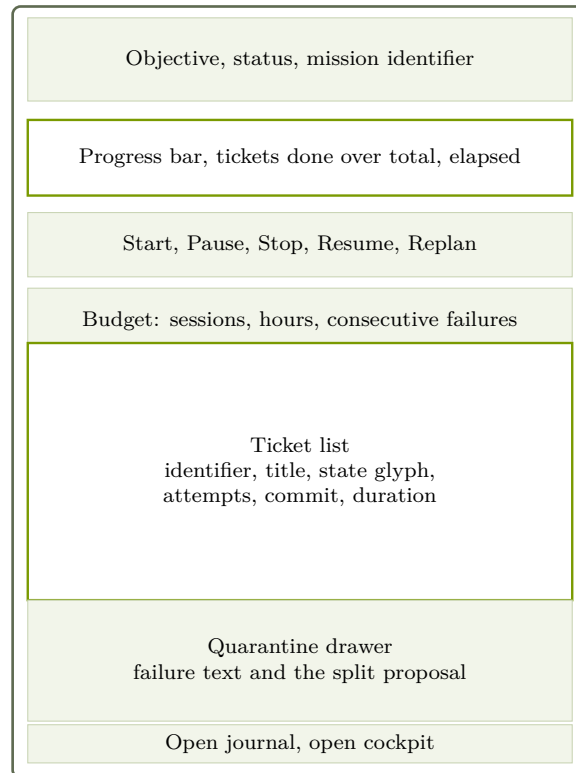


Figure 6.2: Mission Control layout.

Control	Label	Behaviour
ctl.mission.objective	Objective field	Multi line, editable before start, read only during a run
ctl.mission.plan	Plan	Produces the ticket graph without executing, so it can be reviewed
ctl.mission.start	Start	Plans if needed, then runs to completion
ctl.mission.pause	Pause	Halts after the ticket in flight, state preserved
ctl.mission.resume	Resume	Continues from durable state, also available after a crash
ctl.mission.stop	Stop	Halts and marks the mission stopped
ctl.mission.background	Run in background	Detaches the supervisor so it survives closing the window
ctl.mission.budget	Budget	Popover with sessions, hours, attempts per ticket, consecutive failures
ctl.mission.ticket	Ticket row	Click opens the ticket detail: brief, diff, verification output, transcript
ctl.mission.retry	Retry ticket	Requeues a quarantined ticket, optionally with a different vendor
ctl.mission.split	Split ticket	Sends the ticket to the replanner and inserts the children
ctl.mission.skip	Skip ticket	Marks it abandoned and unblocks its dependants where possible

Control	Label	Behaviour
ctl.mission.edit	Edit ticket	Modifies title, scope and files before it runs
ctl.mission.reorder	Drag handle	Reorders within the constraints of the dependency graph
ctl.mission.addticket	Add ticket	Inserts a manual ticket into the graph
ctl.mission.journal	Open journal	Opens the decision journal filtered to this mission
ctl.mission.cockpit	Open cockpit	Opens the end of mission summary and the approval gate

Table 6.6: Mission Control controls.

6.6.1 Ticket states

State	Glyph	Meaning
pending	Hollow circle	Waiting for its dependencies
ready	Outlined circle	Dependencies satisfied, queued
running	Filled ring, animated	A session is executing it now
verifying	Half filled ring	Session finished, tests executing in a snapshot
done	Check	Verified and committed, commit hash shown
retry	Rotating arrow	Failed, queued again with a different vendor
split	Branch glyph	Replaced by children
quarantined	Warning triangle	Exhausted its policy, needs a human
skipped	Dash	Abandoned deliberately

Table 6.7: Ticket state vocabulary. Every state has a glyph, a colour and a word.

6.7 Agent Theatre

Control	Behaviour
ctl.theatre.lane	Expands to the lane's transcript, tool calls and produced artefacts
ctl.theatre.artifact	Opens the artefact: a diff, a verdict, a screenshot, a report
ctl.theatre.kill	Terminates the session in that lane and marks its ticket failed
ctl.theatre.vendor	Switches the vendor for that role for subsequent sessions
ctl.theatre.follow	Follows the active lane automatically as work moves between roles
ctl.theatre.cost	Toggles the cost column between tokens, requests and elapsed
ctl.theatre.graph	Switches from lanes to a node graph showing which role handed work to which
ctl.theatre.filter	Filters lanes by role or by ticket

Table 6.8: Agent Theatre controls.

6.8 Context Inspector

Agent Theatre: live lanes, artefacts, cost				
Planner	idle	last: ticket graph, 19 nodes		
Implementer	running ticket 7	42s	8.1k tokens	
Reviewer	queued	lenses: correctness, security, conventions		
Verifier	running	snapshot tests	14/38 passed	
Repairer	idle	last: fixed a null guard in ticket 5		
Scribe	idle	last: journal entry for ticket 6		

Figure 6.3: Agent Theatre. One lane per role from Table 5.1, each showing its current act and its last artefact. Architect and Replanner lanes appear only while those roles are active, so an idle mission shows six lanes rather than eight.

Control	Behaviour
ctl.context.budget	Budget bar: used versus available, segmented by tier, with the token count
ctl.context.item	One row per included item: path, tier, tokens, why it was included
ctl.context.why	Explains the inclusion reason: structural closure, pinned, similarity fallback, skill
ctl.context.pin	Pins an item so the solver must always include it
ctl.context.exclude	Excludes an item for this workspace and writes the rule to settings
ctl.context.expand	Raises an item's tier, for example from signatures to full body
ctl.context.collapse	Lowers an item's tier to reclaim budget
ctl.context.heat	Heat map over the file tree, coloured by accumulated agent attention
ctl.context.skills	Lists the skills injected into the current session with their source
ctl.context.integrations	Shows which expansion integrations contributed, with a capability toggle each
ctl.context.compression	Shows the compression ratio and the escalation rate for the session
ctl.context.export	Exports the exact assembled context for reproduction and debugging

Table 6.9: Context Inspector controls.

6.9 Verification Deck

Control	Behaviour
ctl.verify.run	Verify now: snapshots the change set and runs the mapped tests
ctl.verify.full	Full suite: runs everything rather than the mapped subset
ctl.verify.result	A run: command, exit code, duration, failing tests, output tail
ctl.verify.rerun	Reruns a single failing test in isolation
ctl.verify.repair	Hands the failure to the repairer inside the snapshot
ctl.verify.visual	Renders the configured page and returns a visual verdict
ctl.verify.compare	Compares the render against a design export
ctl.verify.qa	Runs an adversarial sweep against the running application
ctl.verify.flags	Security and cost flags with severity, file and line
ctl.verify.policy	Opens the verification policy for this repository
ctl.verify.history	Previous runs with their verdicts, filterable by ticket

Table 6.10: Verification Deck controls.

6.10 Atlas

Control	Behaviour
ctl.atlas.mode	Switches between topology, impact, ownership and history views
ctl.atlas.filter	Filters nodes by path, language or module
ctl.atlas.focus	Focuses a node and dims everything outside its neighbourhood
ctl.atlas.depth	Sets transitive depth for impact, from one to unbounded
ctl.atlas.open	Opens the file at the definition
ctl.atlas.mission	Creates a mission scoped to the current selection
ctl.atlas.fleet	Switches the scope from this repository to the registered fleet
ctl.atlas.export	Exports the current view as a vector image for documentation

Table 6.11: Atlas controls.

6.11 Status bar

Control	Behaviour
ctl.status.panels	Labelled toggles: Files, Search, Git, Mission, Agents, Context, Verify, Terminal
ctl.status.diag	Errors and warnings, click opens Problems
ctl.status.verify	Last verification verdict with a glyph, click opens the Deck
ctl.status.tokens	Tokens used in the current session and the budget remaining
ctl.status.cost	Session cost when the provider reports it, otherwise request count
ctl.status.ekg	Activity pulse driven by real agent state: idle, thinking, verifying, failing
ctl.status.position	Cursor position, selection size, indentation, encoding, language

Table 6.12: Status bar controls.

6.12 Command palette and decisions

The palette is the universal entry point and is required to reach every command in this document. It supports three modes: commands, symbols with a leading hash, and files with a leading slash. Missions are startable directly from it with a leading exclamation mark.

Decision cards replace terminal questions. When an agent reaches a genuine fork it emits a choice, and the interface renders two or three cards side by side, each carrying the option name, its measured cost, its structural consequence and a code sketch. Selecting a card records the decision in the journal with its rationale and resumes the mission.

6.13 First run

1. Welcome: product statement, open a project, clone from a provider.
2. Model routing: choose the implementer, the reviewer, the planner and the embedding model, with a local only option offered explicitly.
3. Sign in to the chosen providers, each with its own state indicator.
4. Index the project, showing determinate progress and what is being built.
5. Optional fleet registration.
6. A guided first mission on a scoped objective, so the developer sees the whole loop before trusting it with real work.

6.14 Keymap

Binding	Command
mod+shift+p	Command palette
mod+p	Files
mod+shift+f	Search across the project
mod+i	Inline agent prompt at the cursor
mod+m	Toggle Mission Control
mod+shift+m	Start or resume the mission
mod+alt+m	Stop the mission
mod+g	Toggle Agent Theatre
mod+k then c	Toggle Context Inspector
mod+shift+v	Verify now
mod+alt+v	Full verification
mod+shift+a	Toggle Atlas
mod+alt+i	Impact for the symbol under the cursor
mod+j	Toggle terminal
mod+enter	Accept the agent range under the cursor
mod+backspace	Revert the agent range under the cursor

Table 6.13: Default bindings. Every one is rebindable and every command is reachable without one.

6.15 States that must be designed, not improvised

State	Requirement
Empty project	Explains what to do next with a primary action, never a blank pane
Indexing	Determinate progress, the product remains usable, agents wait rather than guess
No model configured	Blocks agent surfaces with a direct route to routing settings
Provider unauthenticated	Inline sign in with the exact command, never a silent failure
Mission running	Every surface shows it, and closing the window does not kill it
Verification failing	Failure is the loudest thing on screen, with the exact command to reproduce
Quarantined ticket	Failure text and the proposed split, with one click to accept the split
Offline	Local capabilities keep working and remote ones are visibly disabled
Large repository	Progressive indexing, structural queries available before embeddings finish
Conflicting edits	The developer's edit always wins, the agent range is marked stale and reverifies

Table 6.14: Designed states.

Chapter 7

User Stories

Every story below is written from a persona, states the value, and carries acceptance criteria that an implementation can be tested against. Each story names the surfaces from Chapter 6 it exercises, which is what keeps the interface specification honest: a surface with no story behind it should not exist.

7.1 Personas

Persona	Context and pressure
Solo founder	Owens the entire stack, works in bursts, cannot afford a wrong autonomous change reaching production
Staff engineer	Large existing codebase, cares about blast radius and review quality more than raw speed
Team lead	Wants the team's standards enforced mechanically rather than in review comments
Contractor	Enters unfamiliar repositories constantly, needs orientation before change
Regulated team	Must show what changed, who decided it, what verified it, and what left the machine

Table 7.1: Personas driving the specification.

7.2 Orientation and understanding

S-01 Understand an unfamiliar repository

As a contractor, I open a repository I have never seen and want a structural orientation rather than a folder listing.

- Atlas opens on the topology view with modules grouped and entry points marked.
- Structural queries are answerable before embedding indexing completes.
- Selecting any node shows what it imports, what imports it, and the symbols it defines.
- Surfaces: Atlas, Workbench.

S-02 Know the blast radius before changing a shared contract

As a staff engineer, before I rename a function used across the codebase I want the exact set of files that will break, not a guess.

- Impact for the symbol under the cursor returns the definition, importers by transitive distance, and textual references.
- The result is computed from parsed imports, and the interface states that it is deterministic.
- If a similarity fallback contributed any row, that row is labelled as a guess.
- Surfaces: Workbench inline, Atlas.

S-03 See why a line exists before deleting it

As a staff engineer, when an agent proposes removing code I want the history that explains it.

- Blame is available inline without leaving the buffer.
- The agent's context for that region included the commit message that introduced it.
- The Context Inspector shows history as a contributing integration.
- Surfaces: Workbench gutter, Context Inspector.

7.3 Assisted change**S-04 Make a small change and trust it**

As a solo founder, I ask for a small change and want to see only code that already runs.

- The change is executed against the project's own test command in an isolated snapshot before it is displayed.
- The displayed diff carries the verification verdict, the command and the exit code.
- If verification failed, the failure is shown instead of the diff, with the exact command to reproduce it.
- My working tree was never modified during verification.
- Surfaces: Workbench, Verification Deck.

S-05 Correct an agent range without losing my own edits

As any developer, I edit a file while an agent is working in it.

- My edit always wins; the agent range is marked stale rather than overwritten.
- The stale range is reverified before it can be accepted.
- Accept and revert are available per range, not only per file.
- Surfaces: Workbench gutter and inline controls.

S-06 Get a rival opinion on a risky diff

As a team lead, before merging a change to a payment path I want an adversarial review from a different vendor.

- Review runs three lenses in parallel and returns a structured verdict with severities.
- Repository policy can require extra lenses and a lower blocking threshold for configured paths.
- The verdict is stored in the journal and can be exported for an audit.
- Surfaces: Verification Deck, Agent Theatre.

7.4 Autonomous delivery

S-07 Ship a whole feature while I am away

As a solo founder, I state an objective, leave, and return to finished work.

- The objective is decomposed into a dependency ordered ticket graph I can review before execution.
- Each ticket executes in a fresh session, is verified, and is committed separately.
- No prompt, question or confirmation is required for anything the supervisor can decide.
- Progress is visible at a glance from the title bar without opening a panel.
- Surfaces: Mission Control, Agent Theatre, Verification Deck.

S-08 Survive a failure without losing the run

As a solo founder, one ticket fails repeatedly and I do not want the mission to stop.

- The supervisor retries, then swaps vendor, then splits the ticket into smaller children, then quarantines it.
- Work that does not depend on the quarantined ticket continues.
- The quarantine entry shows the failure text and the proposed split, and accepting the split is one action.
- Surfaces: Mission Control.

S-09 Close the laptop mid mission

As any developer, I close the machine during a long run.

- Mission state is durable and resuming continues from the last completed ticket.
- At most the ticket in flight is lost, and its partial work is discarded rather than committed.
- Resuming requires one action and no re planning.
- Surfaces: Mission Control.

S-10 Stop a mission immediately

As any developer, I decide the mission is going the wrong way.

- Stop halts after the ticket in flight, leaving the repository in a committed state.
- Everything already committed remains, and the journal explains what each commit did.
- Restarting with a corrected objective does not require undoing the good work.
- Surfaces: Mission Control, title bar.

S-11 Cap the cost of autonomy

As a team lead, I need a hard ceiling on an unattended run.

- Sessions, wall clock hours, attempts per ticket and consecutive failures are all capped and editable before and during a run.
- Reaching a cap halts the mission with a stated reason rather than continuing.
- Token and cost usage are visible while the run proceeds.
- Surfaces: Mission Control, status bar.

7.5 Correctness of appearance

S-12 Verify a page actually looks right

As a solo founder, after a styling change I want the machine to look at the result.

- The page is rendered headless, screenshotted, and judged against a stated intent.
- The verdict is structured with issues ranked by severity, so it can drive a fix loop.
- High severity issues trigger an automatic repair attempt before the change is presented.
- Surfaces: Verification Deck.

S-13 Match an intended design

As a team lead, the implementation must match the design export.

- A design image can be attached to a ticket or supplied ad hoc.
- Comparison returns concrete differences: layout shifts, missing elements, wrong colours, spacing.
- Differences become tickets rather than a wall of prose.
- Surfaces: Verification Deck, Mission Control.

7.6 Scale and cross repository work

S-14 Change a contract across repositories

As a staff engineer, an endpoint changes in the backend and every consumer must follow.

- Fleet impact separates the defining repository from consuming ones.
- Execution proceeds in dependency order, verifying each repository independently.
- Related changes are prepared together and linked.
- Surfaces: Atlas in fleet scope, Mission Control.

S-15 Register the fleet without editing configuration

As any developer, I want to pick my repositories from my provider rather than write a manifest.

- Repositories are listed from the authenticated provider.
- Selected repositories that are missing locally are cloned.
- Roles are proposed automatically and remain editable.
- Surfaces: Atlas, first run.

7.7 Control over the machine

S-16 See exactly what the agent is reading

As a staff engineer, when output is wrong I want to know what the model was looking at.

- Every included item is listed with its tier, its token cost and its inclusion reason.

- Items included by similarity are labelled distinctly from structural inclusions.
- Any item can be pinned or excluded, and exclusions persist for the workspace.
- The exact assembled context can be exported and replayed.
- Surfaces: Context Inspector.

S-17 Keep attention off the wrong files

As a staff engineer, generated files dominate attention and produce noise.

- The attention heat map shows which files accumulate agent reads.
- Excluding a file or a glob is one action from the heat map.
- The exclusion applies to future context assembly immediately, without reindexing.
- Surfaces: Context Inspector.

S-18 Decide an architectural fork deliberately

As a team lead, when the agent reaches a real decision I want to choose with costs in front of me.

- The choice is presented as cards, each with measured cost and structural consequence.
- Unblocked tickets continue while the choice is pending.
- The decision and its rationale are recorded in the journal.
- Surfaces: decision cards, Mission Control.

S-19 Return to a past decision and take the other branch

As a staff engineer, a decision three commits back turned out wrong.

- The journal shows decisions on a timeline anchored to commits.
- Branching from a decision point is one action and does not disturb the current tree.
- The alternative can be run as a new mission scoped to that branch.
- Surfaces: Atlas history view, Mission Control.

7.8 Team standards and compliance

S-20 Enforce house standards without review comments

As a team lead, every agent session in this repository must follow our conventions.

- Workspace skills override user skills of the same name.
- The Context Inspector shows which skills were injected and from where.
- A violation is caught by the reviewer lens rather than by a human in review.
- Surfaces: Context Inspector, Verification Deck.

S-21 Prove what happened for an audit

As a regulated team, I must show what changed, who decided it and what verified it.

- Every ticket has a commit, a verification record and a journal entry.
- Review verdicts are exportable in a machine readable interchange format.
- Secrets are redacted before any content leaves the machine, and the redaction is recorded.
- Surfaces: Verification Deck, journal export.

S-22 Know what leaves the machine

As a regulated team, I must control every outbound integration.

- Each context expansion integration is a capability with an explicit toggle.
- The Context Inspector shows which integrations contributed to a session.
- A fully local configuration is selectable in first run and remains functional.
- Surfaces: Context Inspector, settings.

7.9 Reliability of the tool itself

S-23 Never lose the editor to an agent

As any developer, an agent process hangs.

- The interface stays responsive because agents run out of process.
- An inactivity watchdog terminates the session and marks the ticket failed.
- Editing, search and navigation continue unaffected.
- Surfaces: all.

S-24 Never hang on an interactive command

As any developer, an agent runs a command that asks a question.

- Interactive commands are refused by policy, with the non interactive form stated.
- Long running processes are started detached and their logs are read instead.
- A command producing no output for a bounded period is terminated.
- Surfaces: terminal, Agent Theatre.

S-25 Keep working while a mission runs

As any developer, I want to edit while the supervisor works.

- Verification runs in snapshots, so it never locks the working tree.
- Indexing is bounded so the machine stays usable.
- Conflicts between my edits and agent ranges are surfaced, never silently merged.
- Surfaces: Workbench, Mission Control.

Chapter 8

Work Breakdown and Issue Register

8.1 Categories

Work is partitioned into twelve categories. Each category maps to a crate set from Chapter 3 and owns a slice of the interface from Chapter 6. The partition is chosen so that two categories can be worked in parallel without editing the same files.

Key	Category	Crates	Owns
PLA	Platform	xyra-ui, xyra-protocol, xyra-memory, xyrad, xyra	Window, element model, service wiring, storage, packaging
TXT	Text core	xyra-text	Rope, transactions, undo, selections
SYN	Syntax	xyra-syntax	Parsing, highlighting, structural motion
LNG	Language	xyra-lang	Language servers, diagnostics, completion, refactors
PRJ	Project	xyra-project, xyra-vcs	Worktree, watching, git
TRM	Terminal	xyra-term	Terminal grid, process supervision
CTX	Context engine	xyra-context	Graph, retrieval, compression, budget solver
AGT	Agent layer	xyra-agent, xyra-skills	Sessions, protocols, tools, skills
MIS	Mission	xyra-mission	Ticket graph, supervisor, budgets, journal
VER	Verification	xyra-verify	Sandbox, visual checks, sweeps, flags
FLT	Fleet	xyra-fleet	Multi repository registry and operations
EXT	Extensions	xyra-ext	Extension host, capabilities, marketplace client

Table 8.1: Work categories and their ownership boundaries.

8.2 Dependency between categories

8.3 Issue register

Identifiers are stable and are referenced by the generation prompts in Chapter 10. The Depends column lists issues that must land first.

ID	Issue	Depends
PLA	Platform	

ID	Issue	Depends
PLA-001	Repository skeleton, workspace manifest, lint and format configuration, license allow list check, continuous integration definition	
PLA-002	Vendor the UI framework at a pinned revision with its license intact	PLA-001
PLA-003	Extend the license check to vendored third party trees and generate the aggregated notice file	PLA-002
PLA-004	xyra-ui: element primitives, flex containers, text, icon, scroll area	PLA-002
PLA-005	xyra-ui: theme tokens from Chapter six, light and dark	PLA-004
PLA-006	xyra-ui: button, input, select, checkbox, toggle, slider with all states	PLA-005
PLA-007	xyra-ui: list with virtualisation and keyboard navigation	PLA-004
PLA-008	xyra-ui: table with sortable columns and sticky header	PLA-007
PLA-009	xyra-ui: popover, dropdown, context menu, tooltip	PLA-006
PLA-010	xyra-ui: modal dialog, confirmation, and focus trapping	PLA-009
PLA-011	xyra-ui: toast and notification centre	PLA-009
PLA-012	xyra-ui: progress bar, determinate and indeterminate, and the activity pulse	PLA-006
PLA-013	Window shell: title bar, activity bar, docks, resizable splits	PLA-004
PLA-014	Dock persistence: sizes, visibility and active panel per workspace	PLA-013
PLA-015	xyra-proto: message types, versioning, backward compatible decoding	PLA-001
PLA-016	xyrad service: supervision tree, graceful shutdown, restart policy	PLA-015
PLA-017	Interface to service transport over a local socket with reconnect	PLA-016
PLA-018	Projection dispatch with version stamps and stale frame rejection	PLA-017
PLA-019	Intent routing from surfaces to the core reducer	PLA-018
PLA-020	Settings: layered defaults, user and workspace, live reload	PLA-016
PLA-021	Command registry and the command palette with three modes	PLA-009
PLA-022	Keymap engine, rebinding, conflict detection, chord support	PLA-021
PLA-023	Storage layer: embedded stores, migrations, integrity checks	PLA-016
PLA-024	Crash reporting that is local by default and opt in for upload	PLA-016

ID	Issue	Depends
PLA-025	Packaging for macOS, Linux and Windows with signed artefacts	PLA-013
PLA-026	Update channel with staged rollout and rollback	PLA-025
PLA-027	Accessibility layer: names, roles, focus order, screen reader output	PLA-006
PLA-028	Internationalisation scaffolding with externalised strings	PLA-006
PLA-029	Headless mode for the core service to run without a window	PLA-016
PLA-030	Interface test harness driving projections and asserting intents	PLA-019
TXT	Text core	
TXT-001	Rope buffer with line and character indexing	PLA-001
TXT-002	Edit transactions with grouping and coalescing	TXT-001
TXT-003	Undo and redo history with selection restoration	TXT-002
TXT-004	Multi cursor and multi selection algebra	TXT-002
TXT-005	Word, line, paragraph and bracket motions	TXT-004
TXT-006	Encoding detection and line ending preservation	TXT-001
TXT-007	Large file strategy with chunked loading and disabled features	TXT-001
TXT-008	Search within a buffer, literal and regular expression	TXT-001
TXT-009	Replace with preview and per match acceptance	TXT-008
TXT-010	Soft wrap, virtual whitespace and indentation guides	TXT-001
TXT-011	External change detection and non destructive reload	TXT-001
TXT-012	Buffer transaction fuzz harness	TXT-003
SYN	Syntax	
SYN-001	Parser host with incremental reparsing on edit	TXT-002
SYN-002	Grammar loading and per language configuration	SYN-001
SYN-003	Highlight queries mapped to theme tokens	SYN-002, PLA-005
SYN-004	Bracket matching and auto closing pairs	SYN-001
SYN-005	Structural selection expand and shrink	SYN-001

ID	Issue	Depends
SYN-006	Code folding by syntax node	SYN-001
SYN-007	Indentation rules derived from the grammar	SYN-002
SYN-008	Symbol extraction feeding the context engine	SYN-001
SYN-009	Injection support for embedded languages	SYN-002
SYN-010	Breadcrumb path from the parse tree	SYN-008
LNG	Language	
LNG-001	Language server process supervision with restart back-off	PLA-016
LNG-002	Capability negotiation and per server feature gating	LNG-001
LNG-003	Diagnostics ingestion, deduplication and gutter rendering	LNG-002
LNG-004	Completion with resolve, snippets and sorting	LNG-002
LNG-005	Hover with type information rendered as structured content	LNG-002
LNG-006	Go to definition, declaration, type definition and implementation	LNG-002
LNG-007	Find references with a grouped result surface	LNG-006
LNG-008	Rename symbol with a preview of every edit	LNG-007
LNG-009	Code actions and quick fixes	LNG-003
LNG-010	Formatting on save and on demand	LNG-002
LNG-011	Signature help	LNG-004
LNG-012	Call hierarchy exposed to the context engine	LNG-006
LNG-013	Workspace symbol search	LNG-002
LNG-014	Inlay hints with a density control	LNG-002
LNG-015	Language server output surface for debugging	LNG-001
PRJ	Project and version control	
PRJ-001	Worktree model with ignore aware traversal	PLA-016
PRJ-002	Filesystem watching with debouncing and rename detection	PRJ-001
PRJ-003	Explorer tree with keyboard navigation and multi select	PRJ-001, PLA-007
PRJ-004	File operations: create, rename, move, delete with undo where possible	PRJ-003

ID	Issue	Depends
PRJ-005	Project wide search with structural filters	PRJ-001
PRJ-006	Search and replace across files with per match acceptance	PRJ-005
PRJ-007	Git status, staging and unstaging by hunk	PRJ-001
PRJ-008	Commit surface with message assistance and amend	PRJ-007
PRJ-009	Branch list, switch, create and delete	PRJ-007
PRJ-010	Diff view, split and unified, with word level highlighting	PRJ-007
PRJ-011	Blame with commit detail on hover	PRJ-007
PRJ-012	Conflict resolution surface	PRJ-010
PRJ-013	Stash operations	PRJ-007
PRJ-014	Remote operations: fetch, pull, push with authentication handling	PRJ-009
PRJ-015	History view feeding the Atlas history mode	PRJ-011
TRM	Terminal	
TRM-001	Terminal grid and escape sequence handling	PLA-004
TRM-002	Process supervision with environment construction	PLA-016
TRM-003	Multiple terminals with tabs and splits	TRM-001
TRM-004	Scrollback, search and copy semantics	TRM-001
TRM-005	Task runner definitions and execution	TRM-002
TRM-006	Output capture available to agents as context	TRM-002, CTX-020
TRM-007	Non interactive policy enforcement on agent invoked commands	TRM-002
TRM-008	Detached process management for servers with log tailing	TRM-002
CTX	Context engine	
CTX-001	Import parsing for TypeScript and JavaScript	SYN-008
CTX-002	Import parsing for Python	SYN-008
CTX-003	Import parsing for Rust	SYN-008
CTX-004	Import parsing for Go and Java	SYN-008
CTX-005	Import resolution to concrete paths with alias support	CTX-001
CTX-006	Symbol table with kind, visibility and location	SYN-008
CTX-007	Call edge extraction per symbol	SYN-008
CTX-008	Graph storage with incremental update on file change	CTX-005, PRJ-002

ID	Issue	Depends
CTX-009	Impact closure query with transitive distance	CTX-008
CTX-010	Neighbourhood query for a single file	CTX-008
CTX-011	Test to source mapping from imports	CTX-008
CTX-012	Test to source mapping refined by execution traces	CTX-011, VER-004
CTX-013	SCP record generation at levels zero, one and two	SYN-008
CTX-014	SCP body handles with content addressed retrieval	CTX-013
CTX-015	Token counting with the target model tokeniser	PLA-001
CTX-016	Budget solver with value, cost, redundancy and dependency	CTX-015, CTX-013
CTX-017	Inclusion floors that cannot be evicted by the solver	CTX-016
CTX-018	Delta encoding dictionary per mission	CTX-014
CTX-019	Stable emission order for provider prefix cache alignment	CTX-016
CTX-020	Context assembly pipeline end to end	CTX-019, CTX-018
CTX-021	Chunking strategy aligned to syntax nodes	SYN-008
CTX-022	Local embedding generation with incremental content hashing	CTX-021
CTX-023	Vector index, quantised and memory mapped	CTX-022
CTX-024	Similarity search labelled as a fallback in results	CTX-023
CTX-025	Local cross encoder reranking of the candidate set	CTX-024
CTX-026	Speculative prefetch driven by graph prediction	CTX-009
CTX-027	Attention accounting per file for the heat map	CTX-020
CTX-028	Context export and replay for debugging	CTX-020
CTX-029	Compression and escalation metrics with a reporting surface	CTX-020
CTX-030	Integration: version history as context	PRJ-011
CTX-031	Integration: language server hover and hierarchy as context	LNG-012
CTX-032	Integration: documentation fetch by library and version	CTX-020

ID	Issue	Depends
CTX-033	Integration: issue tracker linkage	CTX-020
CTX-034	Integration: runtime traces and logs	TRM-006
CTX-035	Integration: database schema and pending migrations	CTX-020
CTX-036	Integration: design assets and tokens	CTX-020
CTX-037	Integration: dependency registry metadata and advisories	CTX-020
CTX-038	Capability toggles and provenance display for every integration	CTX-020
CTX-039	Secret redaction before any context leaves the machine	CTX-020
CTX-040	Context evaluation harness measuring first attempt success	CTX-029
AGT	Agent layer	
AGT-001	Agent process supervision with environment construction	PLA-016
AGT-002	Agent client protocol transport and session lifecycle	AGT-001
AGT-003	Model context protocol client for external tool servers	AGT-001
AGT-004	Tool surface exposure with capability gating	AGT-002
AGT-005	Tool: read file at a requested tier	CTX-013
AGT-006	Tool: structural search with labelled fallback	CTX-024
AGT-007	Tool: impact and graph queries	CTX-009
AGT-008	Tool: edit as a reversible transaction	TXT-002
AGT-009	Tool: run tests inside a snapshot	VER-002
AGT-010	Tool: verify the current change set	VER-003
AGT-011	Tool: render check and design comparison	VER-006
AGT-012	Tool: quality sweep	VER-008
AGT-013	Tool: fleet search and impact	FLT-003
AGT-014	Tool: journal a decision	MIS-014
AGT-015	Tool: escalate a decision to the developer	MIS-016
AGT-016	Transcript capture streamed to disk	AGT-002

ID	Issue	Depends
AGT-017	Inactivity watchdog and hard kill path	AGT-001
AGT-018	Role definitions with distinct mandates and tool grants	AGT-004
AGT-019	Vendor routing per role with a runtime switch	AGT-018
AGT-020	Skill registry with workspace, user and bundled precedence	PLA-020
AGT-021	Skill injection into sessions with provenance recorded	AGT-020
AGT-022	Bundled skills: autonomy, terminal, conventions, review, safety, shipping	AGT-021
AGT-023	Inline agent prompt at the cursor	AGT-002
AGT-024	Agent range tracking with accept and revert	TXT-002, AGT-023
AGT-025	Stale range detection when the developer edits concurrently	AGT-024
AGT-026	Council: parallel lens execution with structured verdicts	AGT-019
AGT-027	Council: policy as code per repository and per path	PLA-020
AGT-028	Council: consensus modes across multiple reviewers	AGT-026
AGT-029	Verdict cache keyed by content hash	AGT-026
AGT-030	Interchange export of verdicts for external scanning	AGT-026
MIS Mission		
MIS-001	Ticket schema, graph storage and atomic state persistence	PLA-023
MIS-002	Planner invocation producing a dependency ordered graph	AGT-018
MIS-003	Plan review surface before execution	MIS-002
MIS-004	Ready ticket selection respecting dependencies	MIS-001
MIS-005	Fresh session execution per ticket with a compiled brief	AGT-002, CTX-020
MIS-006	Verification gate before a ticket counts as done	VER-003
MIS-007	Commit per verified ticket with a structured message	PRJ-008
MIS-008	Retry policy with attempt accounting	MIS-005
MIS-009	Vendor swap on retry	AGT-019
MIS-010	Replanner producing child tickets with bounded depth	MIS-002
MIS-011	Quarantine with failure capture and continuation	MIS-008
MIS-012	Working tree reset between attempts that spares supervisor state	PRJ-007
MIS-013	Budget enforcement across sessions, time, attempts and failures	MIS-001
MIS-014	Decision journal anchored to commits	MIS-007

ID	Issue	Depends
MIS-015	Stop, pause and resume with durable state	MIS-001
MIS-016	Escalation as structured decision cards	PLA-010
MIS-017	Background supervisor surviving interface restarts	PLA-029
MIS-018	Crash resume from durable state	MIS-001
MIS-019	Event stream published for the Theatre and the status bar	PLA-018
MIS-020	Mission Control surface with every control from Chapter six	PLA-013
MIS-021	Ticket detail: brief, diff, verification output, transcript	MIS-020
MIS-022	Manual ticket editing, insertion, reordering and skipping	MIS-020
MIS-023	End of mission cockpit summary with an explicit approval gate	MIS-014
MIS-024	Mission templates for recurring objectives	MIS-002
VER	Verification	
VER-001	Runner detection across common project types	PRJ-001
VER-002	Isolated snapshot execution in a throwaway worktree	PRJ-007
VER-003	Verification verdict with command, exit code and output tail	VER-002
VER-004	Targeted test selection from the test mapping	CTX-011
VER-005	Warm worktree pool to remove setup cost	VER-002
VER-006	Headless render and screenshot capture	PLA-016
VER-007	Visual verdict with severity ranked issues	VER-006
VER-008	Adversarial application sweep with a crawler and input generation	VER-006
VER-009	Accessibility auditing during the sweep	VER-008
VER-010	Defect deduplication and severity ranking	VER-008
VER-011	Repair loop that fixes inside the snapshot and reverifies	MIS-008
VER-012	Security pattern flags with file and line	PRJ-001
VER-013	Cost pattern flags for expensive constructs	PRJ-001
VER-014	Verification Deck surface with every control from Chapter six	PLA-013
VER-015	Verification history filterable by ticket	MIS-021
VER-016	Flake detection across repeated runs	VER-003
FLT	Fleet	

ID	Issue	Depends
FLT-001	Repository registry with roles and local paths	PLA-020
FLT-002	Provider listing and cloning of selected repositories	FLT-001
FLT-003	Cross repository search	CTX-008
FLT-004	Cross repository impact separating definition from consumption	FLT-003
FLT-005	Sequenced execution across repositories with independent verification	MIS-005
FLT-006	Linked change preparation across repositories	PRJ-014
FLT-007	Fleet scope in Atlas	FLT-004
EXT	Extensions	
EXT-001	Extension manifest schema and validation	PLA-020
EXT-002	Sandboxed execution host with a capability list	PLA-016
EXT-003	Host interface: commands, surfaces, settings contributions	EXT-002
EXT-004	Grammar and language contributions	SYN-002
EXT-005	Theme contributions	PLA-005
EXT-006	Tool contributions exposed to agents under capability control	AGT-004
EXT-007	Context integration contributions	CTX-038
EXT-008	Local installation, enable, disable and removal	EXT-001
EXT-009	Registry client with signature verification	EXT-008
EXT-010	Failure isolation so an extension trap never affects the editor	EXT-002

Table 8.2: Issue register. Identifiers are stable and referenced by the generation prompts.

8.4 Milestone composition

Milestones are defined by capability, not by date. A milestone is complete when every issue in it passes its acceptance criteria and the stories it claims are demonstrable end to end.

Milestone	Capability delivered	Stories demonstrable
M0 Shell	Window, docks, palette, settings, service transport, storage	None yet, foundation only
M1 Editor	Open, edit, save, syntax, search, git status and diff	S-01 partially, S-03
M2 Intelligence	Language servers, navigation, diagnostics, refactors	S-01, S-02 partially
M3 Context	Graph, SCP, budget solver, retrieval, inspector	S-02, S-16, S-17

Milestone	Capability delivered	Stories demonstrable
M4 Agent	Sessions, tools, inline prompt, ranges, council	S-04, S-05, S-06, S-20
M5 Verification	Sandbox, targeted tests, visual checks, sweeps, deck	S-04, S-12, S-13
M6 Mission	Supervisor, policy, budgets, journal, Mission Control	S-07 to S-11, S-18, S-19
M7 Fleet	Registry, cross repository impact and execution	S-14, S-15
M8 Platform	Packaging, updates, accessibility, extensions	S-21, S-22, S-23 to S-25

Table 8.3: Milestones defined by demonstrable capability.

8.5 Definition of done

An issue is done when all of the following hold. This list is deliberately uniform so that it can be applied mechanically by a reviewer or by an automated gate.

1. The behaviour matches this document, and any deviation is recorded as an amendment to this document rather than left implicit in code.
2. Tests exist at the appropriate level and pass, including at least one test for the failure path.
3. No control is reachable that is not specified in Chapter 6.
4. Every long running operation has a budget and a cancellation path.
5. No new dependency was added outside the license allow list.
6. The interface remains responsive under the operation, verified by the harness.
7. Accessibility names and keyboard reachability exist for every new control.

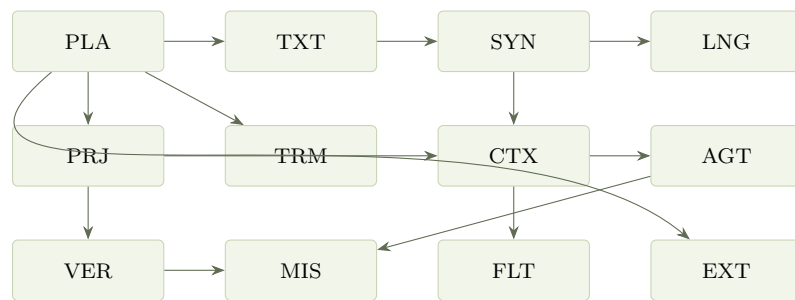


Figure 8.1: Category dependencies. Anything not connected can proceed in parallel.

Chapter 9

Quality, Security and Delivery

9.1 Testing strategy

Level	What it covers	Gate
Unit	Pure logic: rope operations, solver, graph queries, policy decisions	Every commit
Property	Invariants under random input: edits, undo, selections, graph consistency	Every commit
Golden	Compression output, SCP records, context assembly for fixture repositories	Every commit
Integration	Service and interface across the transport, with real stores	Every commit
Interface	Surfaces driven by projections, asserting emitted intents, no window	Every commit
End to end	A real mission on a fixture repository with a scripted agent	Every merge
Live	A real mission with a real model provider on a sample project	Before release
Soak	A long mission with induced failures and a killed process	Before release

Table 9.1: Test levels and when each gate runs.

Two rules matter more than coverage numbers. Every failure path has a test, because failure handling is the product. And every test that touches an agent uses a scripted agent by default, so the suite is deterministic and free; real providers appear only in the live gate.

9.2 Performance budgets

Budgets are enforced by an automated benchmark that fails the build on regression beyond the stated tolerance.

9.3 Security model

9.3.1 Trust boundaries

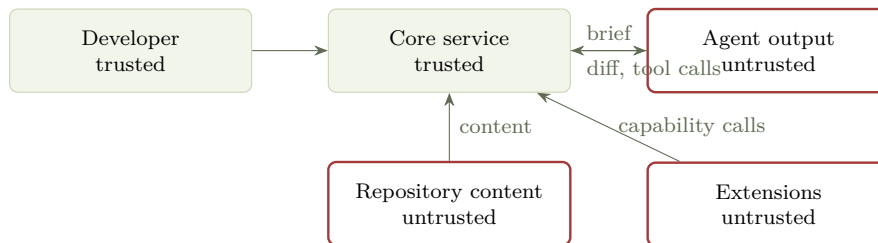


Figure 9.1: Trust boundaries. Everything a model or a repository produces is untrusted input.

9.3.2 Controls

Risk	Control
Prompt injection from repository content	Repository text is data, never instruction. Tool calls originating from content patterns are rejected, and the Context Inspector shows provenance for every item
Secret exfiltration through context	Redaction runs before any content leaves the machine, covering keys, tokens, environment files and private keys, and the redaction is recorded
Malicious extension	Sandboxed execution with an explicit capability list, no ambient filesystem or network, and a trap disables the extension for the session
Destructive agent action	Write access is a property of the role, snapshots are used for execution, and every mutation is a reversible transaction with a commit boundary
Supply chain	Vendored dependencies at pinned revisions, license allow list enforced in continuous integration, signature verification on registry downloads
Credential handling	Provider credentials live in the platform keychain, are never written to the workspace, and are never included in context
Unbounded resource use	Every agent session and verification run carries a timeout, a token budget and a hard kill path
Data at rest	Workspace state is local. Nothing is uploaded without an explicit capability being enabled

Table 9.3: Security controls mapped to risks.

9.4 Privacy and telemetry

Telemetry is off by default and is never a precondition for any feature. When a developer enables it, the payload is enumerable in the settings surface before it is sent, contains no source code, no file paths and no prompts, and can be inspected in a local log. Crash reports are written locally and uploaded only on an explicit action.

9.5 Accessibility

Accessibility is an acceptance criterion on every interface issue rather than a milestone of its own.

- Every control has an accessible name and role, and every surface has a logical focus order.
- Every command is reachable from the keyboard, and the palette is the guaranteed path.
- Contrast meets the standard threshold in both themes, verified by an automated check in continuous integration.
- Severity is never conveyed by colour alone; it always carries a glyph and a word.
- Motion respects the platform reduced motion preference, including the activity pulse.
- Screen reader output is verified for the Mission Control and Verification Deck flows, because those are the surfaces that report autonomous work.

9.6 Internationalisation

Strings are externalised from the first commit, because retrofitting is expensive. The initial shipping locales are English and Turkish. Locale affects interface strings only; identifiers, log output and journal entries remain in English so that artefacts stay portable across teams.

9.7 Platform support

9.8 Packaging and updates

- Artefacts are produced by continuous integration from a clean checkout, never from a developer machine.
- Every artefact carries a checksum and, where the platform supports it, a signature.
- Updates are staged: a fraction of installations receive a release first, and a regression halts the rollout.
- An update never rewrites workspace state in place without a backup that the previous version can still read.
- An air gapped installation path exists, because enterprise customers require it and it also proves the build has no hidden network dependency.

9.9 Risk register

Risk	Severity	Mitigation
Rendering framework churn upstream	Medium	Vendored and pinned, wrapped behind an internal crate so it can be replaced
Model provider terms change	High	Vendor routing is configurable per role, local models are a first class path, no feature depends on one provider
Autonomous run produces plausible but wrong work	High	Verification gate before display, adversarial review, one commit per ticket, journal, easy revert
Context engine underperforms naive stuffing	Medium	Permanent evaluation harness measuring first attempt success, with the naive strategy as the baseline
Scope creep into a general purpose editor	High	Non goals stated in Chapter one, every issue traceable to a story
Single maintainer bottleneck	High	Category partition designed for parallel work, documentation as the source of truth rather than tribal knowledge
Performance regression as features land	Medium	Budgets enforced by benchmark in continuous integration
Extension ecosystem security incident	Medium	Capability sandbox, signature verification, isolation on trap
License contamination through a transitive dependency	High	Allow list enforced mechanically on every build, vendored tree audited

Risk	Severity	Mitigation
------	----------	------------

Table 9.5: Risk register with mitigations that are implemented rather than aspirational.

Operation	Budget	Measured on
Cold start to an interactive window	Under one second	Reference machine, warm cache
Open a file of one hundred thousand lines	Under two hundred milliseconds	Reference machine
Keystroke to rendered frame	Under sixteen milliseconds at the ninety ninth percentile	Reference machine
Incremental reparsing after an edit	Under five milliseconds for a typical file	Reference machine
Structural impact query	Under fifty milliseconds on a repository of ten thousand files	Warm index
Context assembly for a ticket	Under one second excluding model latency	Warm index
Snapshot creation for verification	Under two seconds with a warm worktree pool	Reference machine
Memory at rest with a large repository indexed	Under one gigabyte	Reference machine

Table 9.2: Performance budgets. A regression beyond tolerance fails the build.

Platform	Status	Notes
macOS arm64	Primary	Reference development and benchmark target
macOS x86_64	Supported	Built from the same source, verified in continuous integration
Linux x86_64	Supported	Distributed as a portable single file bundle
Linux arm64	Best effort	Built, tested less frequently
Windows x86_64	Supported	Installer, path length and process semantics handled explicitly

Table 9.4: Platform support matrix.

Chapter 10

Generation Prompts

This chapter turns the document into code. Each prompt is written to be pasted into an agent session together with this document. They assume the agent can read files, run commands and create commits, and they assume nothing else.

10.1 Rules of engagement

Prepend this to every session. It is the contract that keeps generated work consistent with the specification.

You are implementing Xyra, an agentic IDE, from a written specification.

The specification is authoritative. If your instinct disagrees with it, follow the specification and state the disagreement in your final message rather than silently deviating.

Rules that apply to every session:

1. Implement exactly one issue identifier from the register. Do not start another.
2. Never add a dependency whose license is outside MIT, Apache-2.0, BSD, ISC, Zlib, Unlicense or CCO. If you need one that is not, stop and report it.
3. Write no code comments. Names carry the meaning. Rationale goes in the commit body. Documentation is not a comment: module and item documentation is required where the specification asks for it, and it describes what a thing is rather than why a line exists.
4. Never use an em dash anywhere: not in code, strings, documents or commit messages.
5. Interface copy is sentence case. No capitalised words.
6. Every control you add must already exist in the interface chapter. If it does not, stop and report the gap rather than inventing a control.
7. Every long running operation takes a budget and a cancellation token.
8. Write tests, including at least one for the failure path. Run them. Report the result with the exact command and its exit code.
9. Leave the working tree building and the test suite green. Do not commit; report the diff and let the supervisor commit.
10. Never run an interactive command or a process that does not exit. Use the non interactive flag, or start it detached and read the log.

When you finish, report: the issue identifier, the files changed, the test command and its result, and anything in the specification you found ambiguous.

10.2 Bootstrap prompt

Run once, on an empty repository.

Read the attached blueprint. Create the repository skeleton described in the architecture chapter: a workspace manifest, one directory per crate listed in the crate topology table, a lint and format configuration, a license allow list check, and a continuous integration definition that runs format, lint, test and the license check.

Every crate is created with a doc header stating its single responsibility as written in the crate topology table, and with a placeholder test so the suite is green from the first commit. Do not implement behaviour yet.

This is issue PLA-001. Follow the rules of engagement.

10.3 Category prompts

Each of the twelve categories has a lead prompt that establishes shared understanding before individual issues are executed against it.

10.3.1 Platform

You own the PLA category: the window shell, the element toolkit, the core service and the transport between them.

Read the architecture chapter and the design system section of the interface chapter. Implement the design tokens exactly as tabulated: spacing, radius, control heights, text sizes, icon sizes. No arbitrary values anywhere in the toolkit.

Then implement the controls listed in PLA-004 through PLA-012, each with its full state set: default, hover, active, focused, disabled, loading. Every control gets an accessible name and keyboard reachability. Every control gets an interface test through the harness.

Work issue by issue in the order given in the register. One issue, one report.

10.3.2 Text core

You own the TXT category: the rope buffer and everything that edits it.

Correctness is the entire job here. Implement transactions such that undo restores both content and selection, coalescing is predictable, and concurrent transactions on one buffer are ordered rather than merged optimistically.

Write property tests before implementation: random edit sequences must round trip through undo and redo, and the rope must agree with a naive string model on every operation.

Start with TXT-001 and proceed in register order.

10.3.3 Syntax

You own the SYN category: parsing and everything derived from a parse tree.

The parse tree is the substrate for the context engine, not only for colouring. Symbol extraction in SYN-008 must produce, for every file, the ordered symbol list with full signatures, type declarations, documentation comments and outbound call edges, in the shape the context chapter calls a Structural Context Protocol record.

Design that output format first, write golden tests for it against fixture files in each supported language, then implement.

10.3.4 Language

You own the LNG category: language server integration.

Supervision is as important as the features. A server that crashes restarts with backoff and the editor never stops working. A server that lacks a capability degrades that feature visibly rather than failing silently.

Implement LNG-001 and LNG-002 fully before any feature issue, because every feature depends on correct capability negotiation.

10.3.5 Project and version control

You own the PRJ category: the worktree, watching and git.

Two invariants: the developer's edit always wins over any automated change, and no operation may leave the repository in a state the developer did not ask for. Every destructive operation is reversible or confirmed.

Implement staging by hunk in PRJ-007 with the same data model the diff view uses, so the two surfaces cannot disagree.

10.3.6 Terminal

You own the TRM category: terminal emulation and process supervision.

TRM-007 is the issue that matters most: enforce the non interactive policy on commands invoked by agents. A command that would prompt is refused with the correct non interactive form stated, a process that does not exit is started detached with its log tailed, and a command producing no output for a bounded period is terminated.

Implement TRM-001 and TRM-002 first, then TRM-007 before the remaining features.

10.3.7 Context engine

You own the CTX category: the structural graph, the compression protocol, the budget solver and the retrieval pipeline.

Read the context chapter completely before writing code. Implement in this order: structural extraction, then the SCP record at three levels, then token counting with a real tokeniser, then the budget solver, then delta encoding, then the assembly pipeline. Similarity search comes last and is always labelled as a fallback in its results.

Build the evaluation harness described in CTX-040 early, not at the end. Every subsequent change to the engine must be justified by a measured improvement in first attempt success, compression ratio or escalation rate against the naive baseline.

10.3.8 Agent layer

You own the AGT category: agent sessions, the tool surface and skills.

Every tool has a typed contract, a capability gate and an audit record. No tool grants ambient authority. Write access is a property of the role as tabulated in the agent chapter, not of the model.

AGT-017, the inactivity watchdog and hard kill path, is not optional and is not last. A hung agent process must never be able to block the interface or a mission.

10.3.9 Mission

You own the MIS category: the supervisor that runs a project to completion unattended.

The state machine is specified in the agent chapter. Implement it exactly, including the four step failure escalation: retry, vendor swap, split with bounded depth, quarantine and continue.

Two hazards to handle explicitly. The working tree reset between attempts must spare the supervisor's own state directory, otherwise the supervisor destroys its own memory. And ticket splitting must be depth bounded, otherwise a persistently failing ticket splits forever.

State is written atomically after every transition. Prove resumption with a test that kills the process mid ticket and continues.

10.3.10 Verification

You own the VER category: everything that proves a change works.

Verification never touches the developer's working tree. Snapshot the change set into a throwaway worktree and execute there.

Implement VER-001 through VER-003 first, because the mission category depends on them. Then targeted test selection, then visual verification, then the adversarial sweep.

Every verdict is structured, with severity ranked issues, because verdicts feed automatic fix loops rather than human reading.

10.3.11 Fleet

You own the FLT category: multi repository operations.

Registration comes from the developer's provider, never from a hand written file. Impact analysis must separate the repository that defines a contract from those that consume it, because that ordering is what makes a cross repository change safe.

Each repository is verified independently before any linked change is prepared.

10.3.12 Extensions

You own the EXT category: the extension host.

Extensions are untrusted. No ambient filesystem access, no network unless granted, an explicit capability list in the manifest, and a trap disables the extension for the session without affecting the editor.

Implement EXT-002 and EXT-010 before any contribution point, because isolation is the feature.

10.4 Issue prompt template

For each issue in the register, substitute and run.

Implement issue {ID}: {title}.

Read these sections of the blueprint before writing code:

- the architecture chapter section for {crate}
- the interface chapter section for {surface}, if this issue touches the interface
- the user stories that reference this capability: {story ids}

Acceptance:

- The behaviour matches the specification exactly.
- Tests exist including a failure path test, and they pass.
- No control exists that is not specified in the interface chapter.
- No dependency was added outside the license allow list.
- The operation has a budget and a cancellation path if it can run long.
- Accessible name and keyboard reachability exist for every new control.

Dependencies already landed: {depends}.

Follow the rules of engagement. Report when done.

10.5 Self hosting prompt

The most efficient way to build Xyra is to use the existing Xyra mission supervisor to build it. Once the register is loaded as a ticket graph, the loop in the agent chapter applies to this project as it does to any other.

Convert the issue register in the blueprint into a mission ticket graph. Each issue becomes one ticket: the identifier is the ticket identifier, the issue text is the scope, and the Depends column becomes the dependency list.

Then run the mission to completion with these budgets: attempts per ticket three, split depth one, and a stop on five consecutive failures.

For every ticket, the brief is the issue prompt template filled in from the register, with the rules of engagement prepended.

Verification for this project is the workspace test suite plus the format and lint checks plus the license allow list check. A ticket counts as done only when all four pass in an isolated snapshot.

10.6 Review prompt

Run against every diff before it is accepted, ideally on a different vendor from the one that wrote it.

You are reviewing a diff produced by another vendor against a written specification.

Check, in this order:

1. Correctness. Does it do what the issue says, including the failure path.
2. Specification fidelity. Does any control, message or behaviour deviate from the blueprint. Quote the deviation.
3. Boundaries. Does it call across a layer in the wrong direction, or reach into another category's crate.
4. Security. Untrusted input treated as instruction, secrets in context, missing capability gate, unbounded resource use.
5. Conventions. Comments present, em dashes present, capitalised interface copy, arbitrary spacing values instead of tokens.

Return a structured verdict: severity per finding, file and line, and a blocking decision. Default to blocking when uncertain about correctness or security.

10.7 Specification amendment prompt

The document is the source of truth, which means it must be maintained as the code teaches you things.

While implementing {ID} you found that the blueprint is wrong, ambiguous or incomplete.

Do not silently deviate. Produce an amendment:

- The exact section and sentence that is wrong.
- What the code must do instead, and why the specification could not have anticipated it.
- Any other section that must change for the document to stay internally consistent.
- The stories and issues affected.

Emit the amendment as a patch to the blueprint source alongside the code change. The document and the code land together or not at all.

Appendix A

Traceability

This chapter exists to keep the document honest. Every surface must be justified by a story, every story must be delivered by issues, and every issue must belong to a category with an owner. A row that cannot be completed is a defect in the specification, not a detail to be resolved later.

A.1 Story to issue to surface

Story	Capability	Delivered by	Surface
S-01	Orient in an unknown repository	CTX-008, CTX-010, FLT-007	Atlas
S-02	Blast radius before a change	CTX-009, LNG-012, AGT-007	Workbench, Atlas
S-03	History behind a line	PRJ-011, CTX-030	Workbench, Inspector
S-04	Trust a small change	VER-002, VER-003, MIS-006	Workbench, Deck
S-05	Concurrent human edit	AGT-024, AGT-025	Workbench
S-06	Adversarial review	AGT-026, AGT-027, AGT-028	Deck, Theatre
S-07	Unattended feature delivery	MIS-002 to MIS-007, MIS-020	Mission Control
S-08	Failure without stopping	MIS-008 to MIS-011	Mission Control
S-09	Survive a closed laptop	MIS-001, MIS-018	Mission Control
S-10	Stop cleanly	MIS-015	Mission Control, title bar
S-11	Cap the cost	MIS-013	Mission Control, status bar
S-12	Visual correctness	VER-006, VER-007	Deck
S-13	Match a design	VER-007, AGT-011	Deck, Mission Control
S-14	Cross repository contract change	FLT-004, FLT-005, FLT-006	Atlas, Mission Control
S-15	Register a fleet	FLT-001, FLT-002	Atlas, first run
S-16	See what the agent reads	CTX-020, CTX-028, CTX-038	Inspector
S-17	Redirect attention	CTX-027, CTX-038	Inspector
S-18	Decide a fork deliberately	MIS-016, AGT-015	Decision cards
S-19	Branch from a past decision	MIS-014, PRJ-015	Atlas history
S-20	Enforce standards mechanically	AGT-020, AGT-021, AGT-027	Inspector, Deck
S-21	Prove what happened	MIS-014, AGT-030, VER-015	Deck, journal
S-22	Control what leaves the machine	CTX-038, CTX-039	Inspector, settings
S-23	Agent hang cannot break the editor	AGT-017, PLA-016	All
S-24	No interactive command hangs	TRM-007, TRM-008, AGT-022	Terminal, Theatre

Story	Capability	Delivered by	Surface
S-25	Work while a mission runs	VER-002, VER-005, AGT-025	Workbench, Mission Control

Table A.1: Every story maps to issues and to the surfaces that expose it.

A.2 Surface to category ownership

Surface	Owning category	Contributing categories
Workbench	TXT	SYN, LNG, PRJ, AGT, PLA
Mission Control	MIS	AGT, VER, PLA
Agent Theatre	AGT	MIS, PLA
Context Inspector	CTX	AGT, PLA
Verification Deck	VER	MIS, PLA
Atlas	CTX	PRJ, FLT, PLA
Terminal, inside Workbench	TRM	PLA
Settings and palette	PLA	all

Table A.2: Surface ownership. One category owns each surface and is accountable for it.

Appendix B

Formats and Glossary

B.1 Workspace state layout

.xyra/	
mission.json	durable mission state, atomically replaced
mission.log	supervisor log, append only
journal.jsonl	decisions anchored to commits
fleet.json	registered repositories and roles
policy.toml	verification and review policy for this repository
settings.toml	workspace settings, layered over user settings
skills/	workspace skills, highest precedence
index/	symbol table, import graph, chunk metadata
vectors/	quantised embeddings, memory mapped
cache/	content addressed artefacts, safe to delete
views/	generated report artefacts

Deleting the entire directory must degrade the product to a cold start and must never corrupt the repository. This is a tested invariant, not an intention.

B.2 Ticket schema

```
{
  "id": "MIS-007",
  "title": "Commit per verified ticket",
  "scope": "precise enough to act on without asking a question",
  "files": ["crates/xyra-mission/src/commit.rs"],
  "depends_on": ["MIS-006"],
  "status": "pending | ready | running | verifying | done | retry | split | quarantined | skipped",
  "depth": 0,
  "attempts": 0,
  "vendor_index": 0,
  "commit": null,
  "notes": []
}
```

B.3 Verification verdict schema

```
{
  "ok": true,
  "command": "cargo test --workspace",
  "exit_code": 0,
  "duration_ms": 41230,
  "scope": "targeted | full",
  "failures": [],
  "output_tail": "...",
  "snapshot": "throwaway worktree identifier"
}
```

B.4 Review verdict schema

```
{
  "decision": "clean | approve_with_notes | block | inconclusive",
  "lenses": ["correctness", "security", "conventions"],
  "findings": [
    {
      "severity": "critical | high | medium | low",
      "lens": "security",
      "file": "crates/xyra-agent/src/tools.rs",
      "line": 214,
      "issue": "one sentence statement of the defect",
      "evidence": "concrete inputs or state that produce it"
    }
  ],
  "policy": "path rule that decided the blocking threshold"
}
```

B.5 Glossary

Term	Definition
Accelerator	A mechanism that reduces latency or repeated work without changing which context items are selected
Agent range	A contiguous region of a buffer written by an agent and not yet accepted by a human
Atlas	The structural surface: topology, impact, ownership and history
Budget solver	The component that selects context items under a token limit using value, cost, redundancy and dependency
Capability	An explicitly granted permission for an integration, tool or extension
Council	Adversarial review of a diff by a rival vendor across parallel lenses
Delta encoding	Transmitting only what changed since the previous session, referencing the rest by handle
Escalation rate	The frequency with which an agent must request a higher context tier
Fleet	A registered group of repositories treated as one unit for search, impact and execution
Handle	A content addressed reference to a body that can be expanded on request
Impact closure	The deterministic set of files affected by a change, by transitive distance
Intent	A user action sent from a surface to the core, never a direct mutation
Journal	The append only record of decisions anchored to commits
Lens	One perspective in a review: correctness, security or conventions
Mission	An objective decomposed into a ticket graph and executed to completion
Projection	A serialisable snapshot of state prepared for one surface to render
Quarantine	The state of a ticket that exhausted its failure policy and needs a human

Term	Definition
Role	An agent specialisation with its own mandate, tool grant and write access
SCP	Structural Context Protocol, the compressed structural representation of a file
Skill	A durable instruction set injected into agent sessions, resolved by precedence
Snapshot	A throwaway worktree containing a change set, used for verification
Surface	One of the six primary product views
Ticket	The unit of autonomous work: one session, one verification, one commit
Tier	A level on the context ladder, from repository skeleton to runtime evidence

Table B.1: Glossary of terms used consistently throughout this document.

B.6 Document maintenance

This document is the source of truth for the product. Three rules keep it that way.

1. Code that contradicts the document is a defect in one of them, and the discrepancy is resolved before the code lands, not afterwards.
2. An amendment lands in the same change as the code it describes, using the amendment prompt in Chapter 10.
3. A new surface, control or capability enters through a story first. If no persona needs it, it does not ship.